西南交通大学
Southwest Jiaotong University

SCAI011112 视觉计算

生成式模型

计算机科学与技术专业

2025-2026 (2)

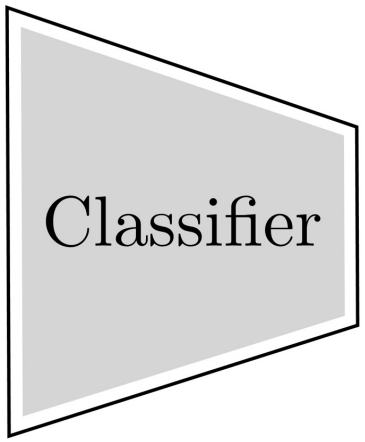
本章主要内容

- **生成式模型概述**
- **变分自编码器（VAE）**
- **扩散模型**
- **基于分数的生成**

判别式与生成式模型



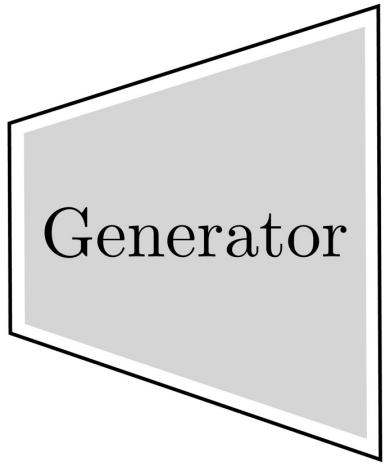
x



Bird
y

分类，
多对一

Bird
y



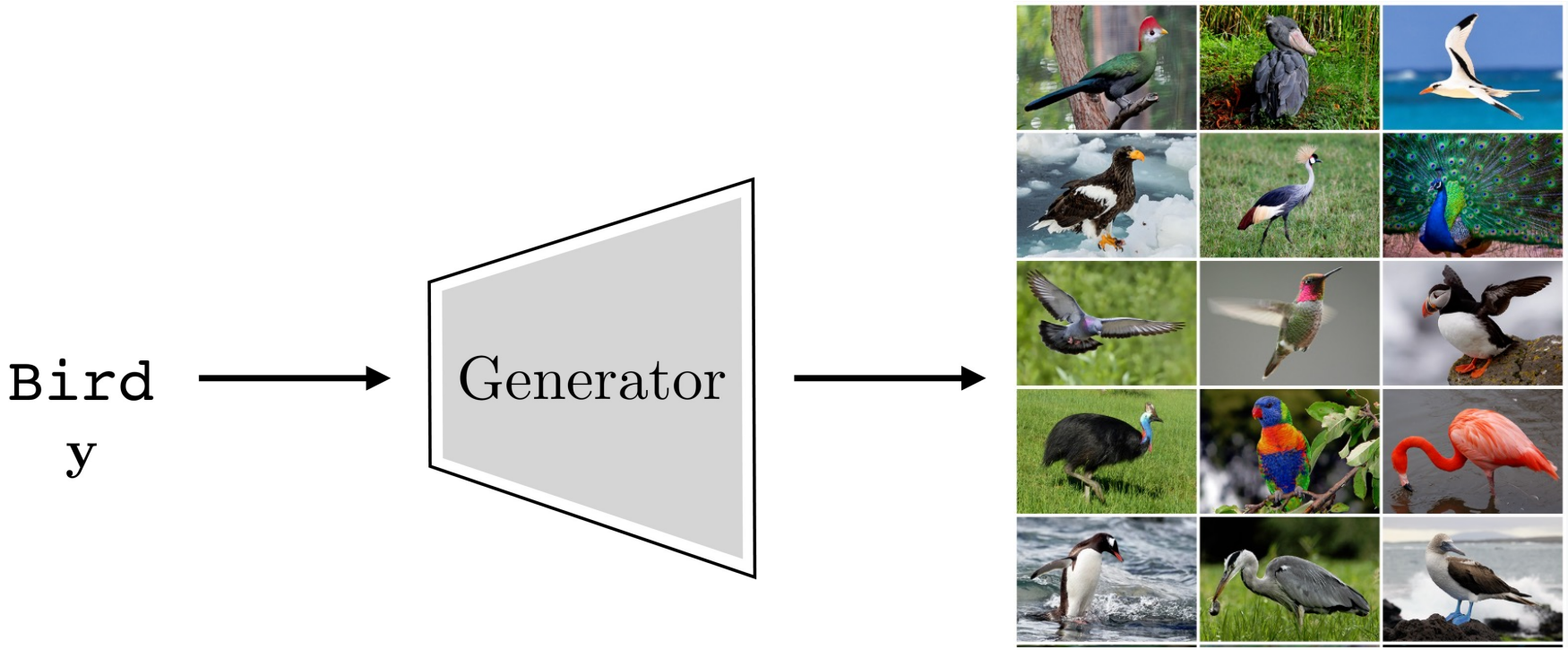


x

生成，
一对多

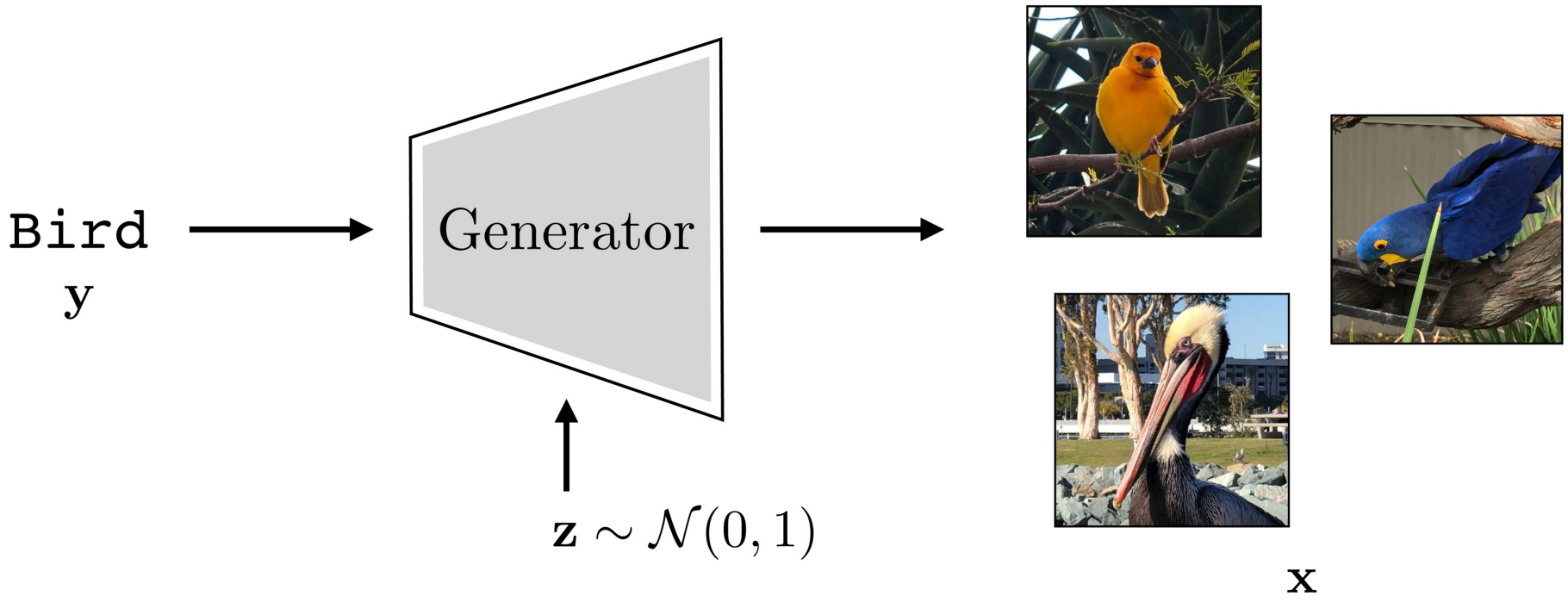
生成式模型

- 对于一个输入，其输出有很多可能的“正确”结果，某些预测结果看上去可能性更大
- 训练集没有严格准确的答案
- 输出空间维度往往很高



生成式模型

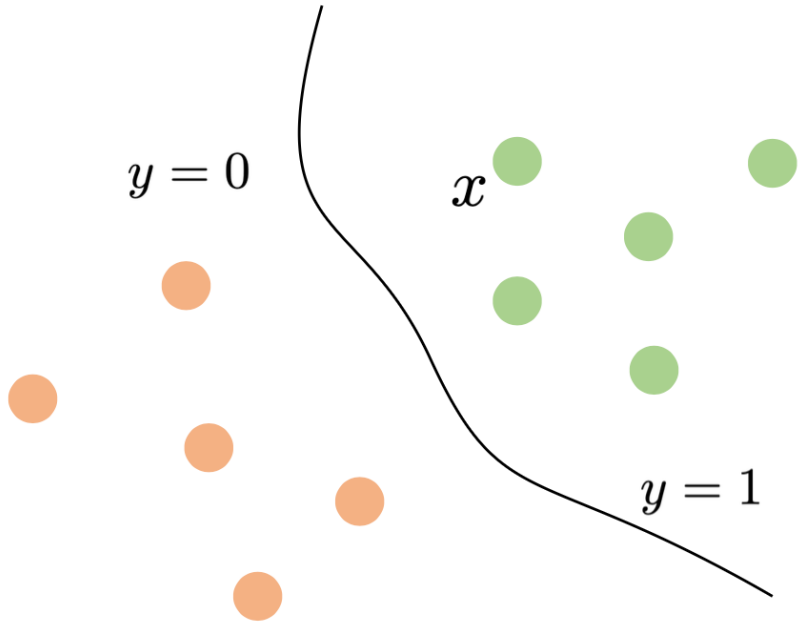
- 对于一个输入，其输出有很多可能的“正确”结果，某些预测结果看上去可能性更大
- 训练集没有严格准确的答案
- 输出空间维度往往很高

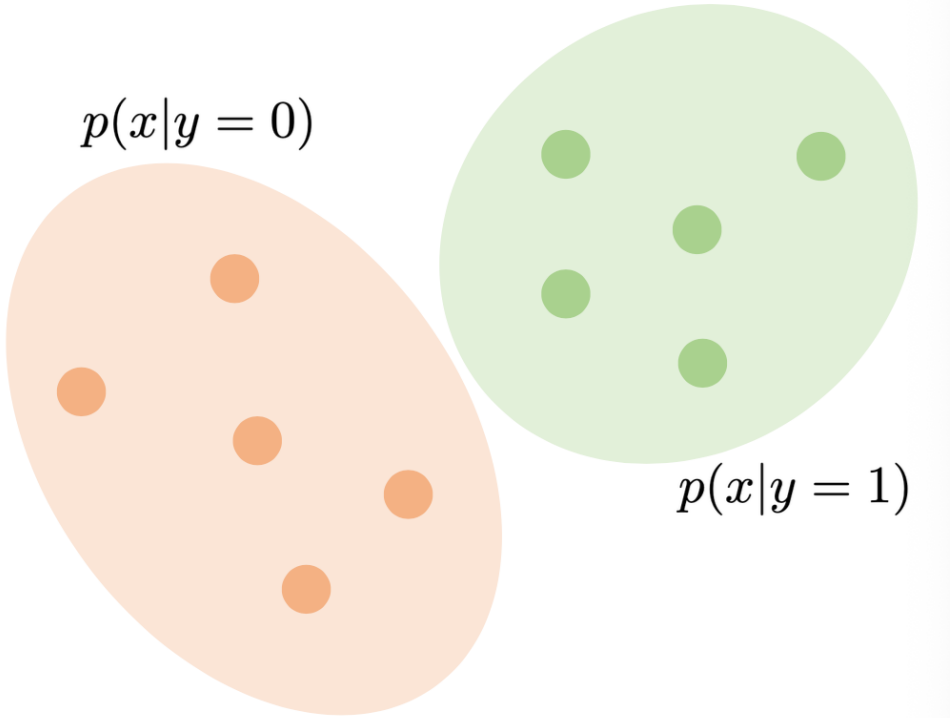


生成式模型

- Usually we draw **z** from a simple distribution such as Gaussian noise
  - $\boldsymbol{p(Z) = \mathcal{N}(0, 1)}$
- The way to think of **z** is that it is a vector of **latent variables**, which specify all the attributes of the images other than those determined by the label
  - **y** tells *g* to make a bird but the **z**-vector is what specifies exactly which bird.
  - Different dimensions of **z** specify different attributes, such as the size, color, pose, background, and so on
- The z-vector is also sometimes called noise. In generative modeling, ***the noise is the signal***; it determines exactly which image to generate.

生成式 VS 判别式模型





判别式模型 P(**Y**|**X**)

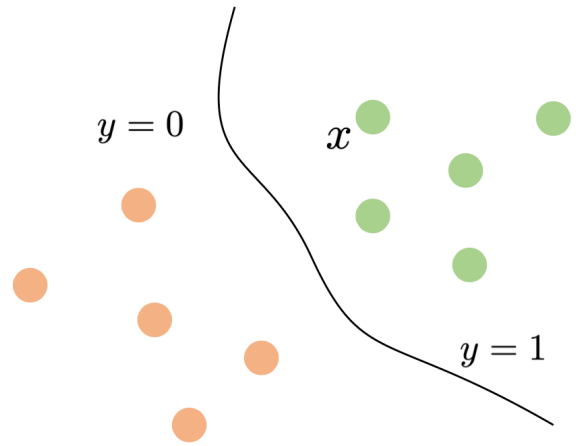
生成式模型 P(**X**|**Y**)

生成式 VS 判别式模型

- **判别式模型**

不分别对p(Y)和P(X|Y)建模

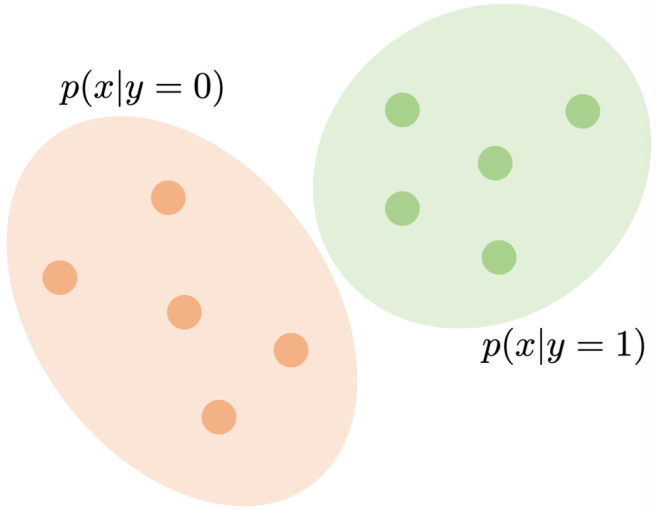
$$\mathbf{P(Y|X;w)} = \frac{\mathbf{P(Y)P(X|Y)}}{\mathbf{P(X)}}$$



只划定事物的边界（判别）

- **生成式模型**
  - 对类条件概率 $P(x \mid y)$ 与先验 $P(y)$ 分别建模

$$\mathbf{P(Y|X;w)} = \frac{\mathbf{P(Y)P(X|Y)}}{\mathbf{P(X)}}$$

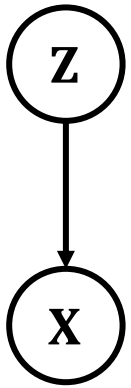


去理解事物诞生的底层规律（生成）

Naive Bayes，GMM，HMM，Bayesian Nets

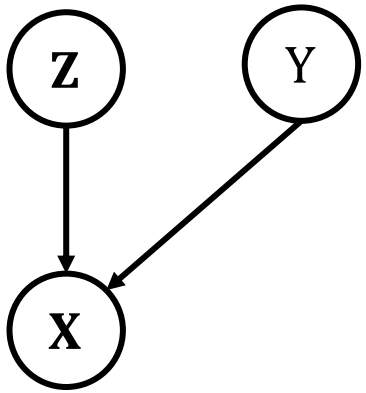
生成式模型

- **非条件生成：** 对 $P(X)$ 进行建模
- **条件生成：** 对 $P(X|Y)$ 进行建模



z~P(Z)

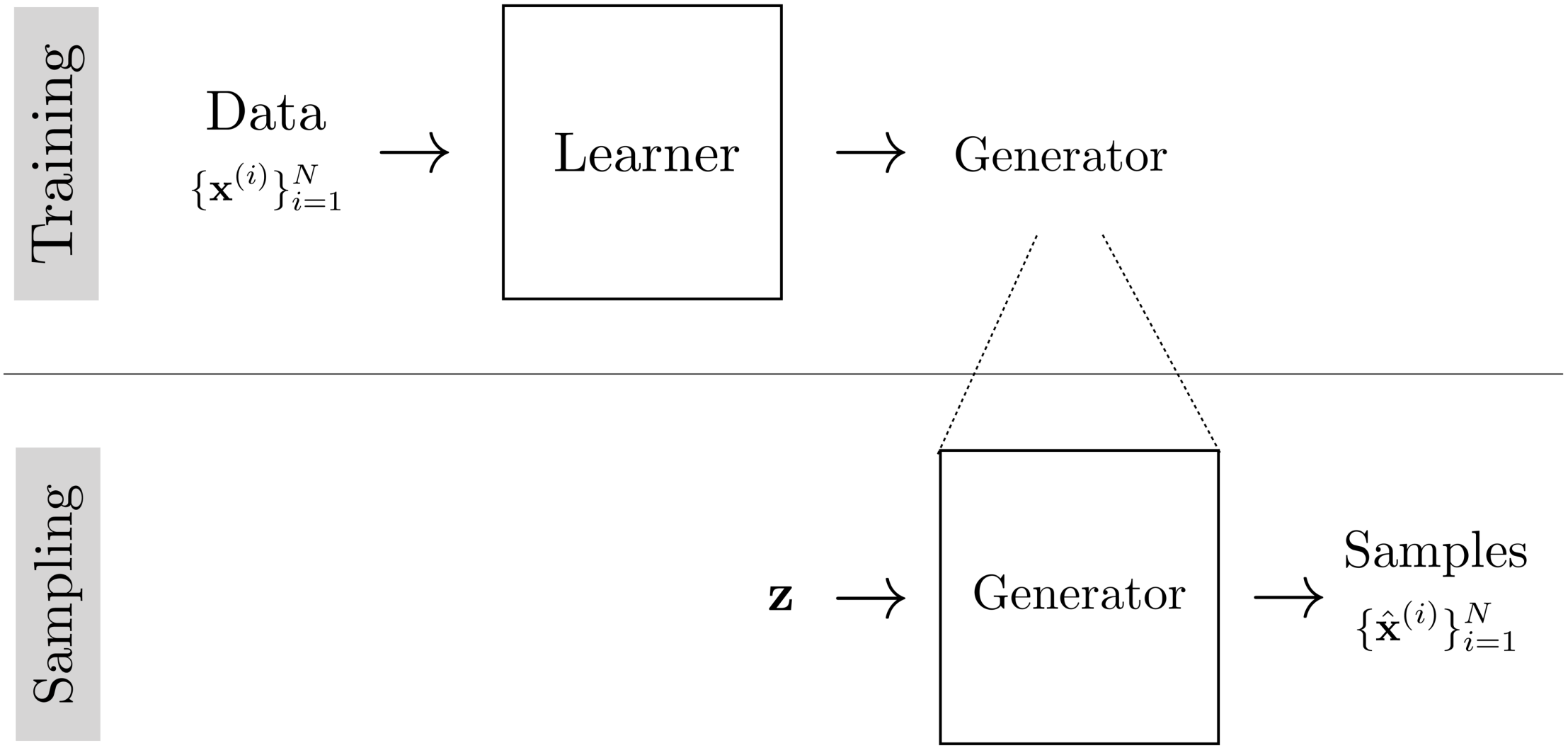
x~g(Z)



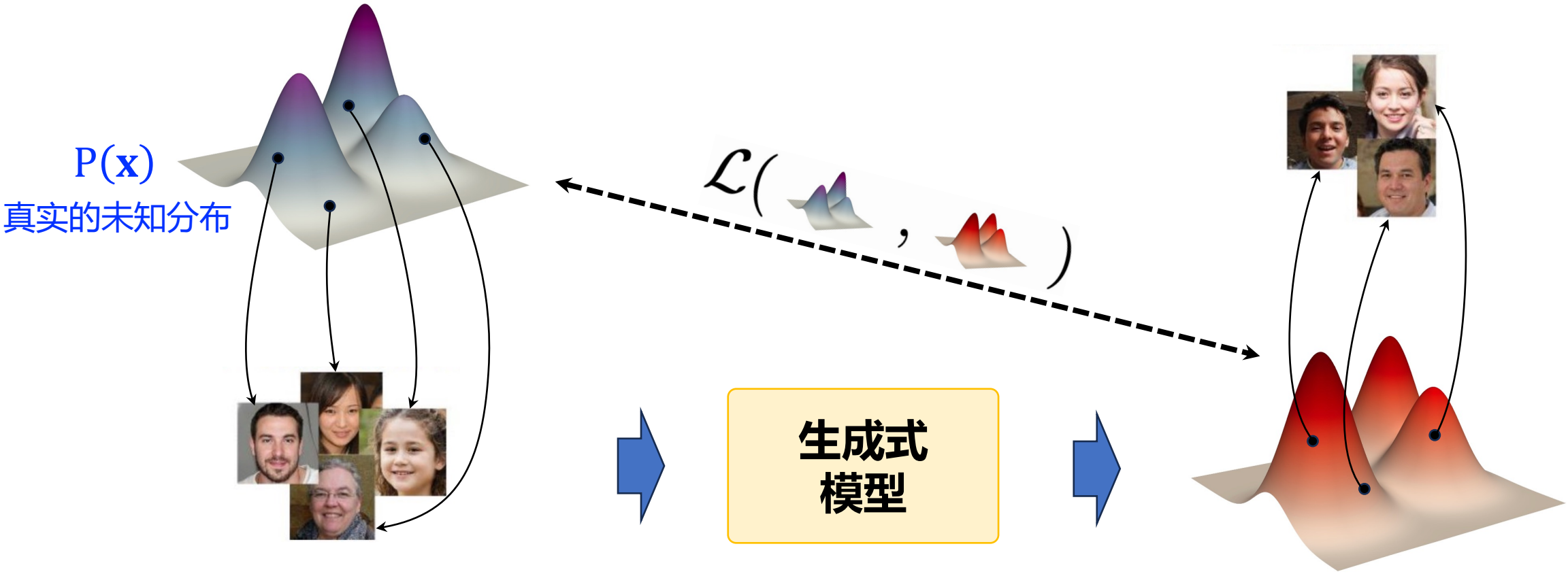
z~P(Z)

x~g(Z, Y)

Learning Generative Models



生成式模型



$$p_D(x) = \frac{1}{N}\sum_{n=1}^{N}\delta(x_n - x)$$

经验分布

$$\min_{\theta} D_{KL}\left(p_D(x) \parallel p_w(x)\right)$$

$p_w(x)$

模型分布

计算 $D_{KL}(p_D(x) \| p_w(x))$

- 最小化KL散度，等价于极大似然

$$D_{KL}(p_D(x) \| p_w(\mathbf{x})) = \mathbb{E}_{p_D(x)}\left[\log \frac{p_D(x)}{p_w(\mathbf{x})}\right]$$

$$= \underbrace{\mathbb{E}_{p_D(x)}[\log p_D(x)]}_{\text{真实样本的熵取负号（常量）}} - \underbrace{\mathbb{E}_{p_D(x)}[\log p_w(\mathbf{x})]}_{\text{对数似然}}$$

从$p_D(x)$中采样，估计该期望

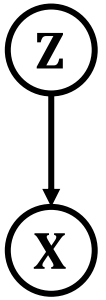
生成式模型

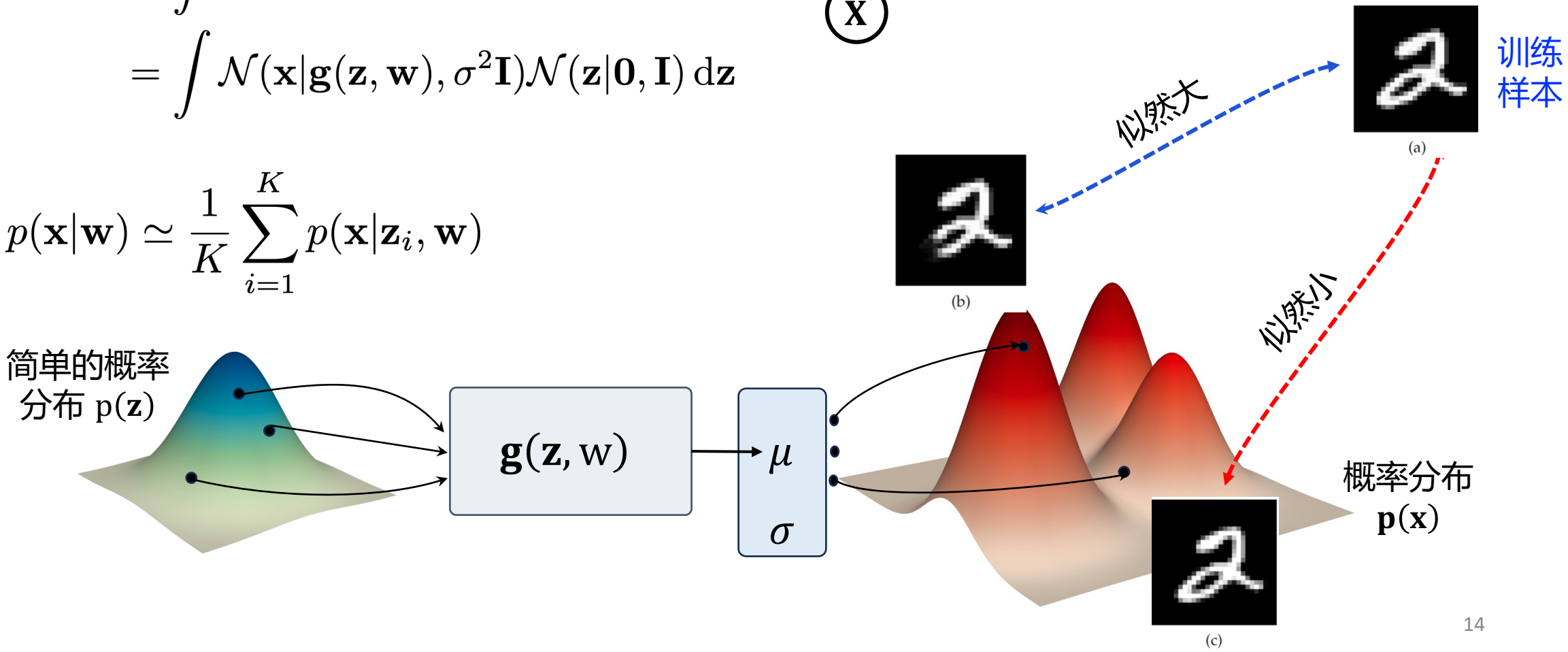
- 目标使 $\mathbb{E}_{p_D(x)}[\log p(\mathbf{x}|w)]$最大

$$p(\mathbf{x}|\mathbf{w}) = \int p(\mathbf{x}|\mathbf{z},\mathbf{w})p(\mathbf{z})\,\mathrm{d}\mathbf{z}$$

$$= \int \mathcal{N}(\mathbf{x}|\mathbf{g}(\mathbf{z},\mathbf{w}),\sigma^2\mathbf{I})\mathcal{N}(\mathbf{z}|\mathbf{0},\mathbf{I})\,\mathrm{d}\mathbf{z}$$

$$p(\mathbf{x}|\mathbf{w}) \simeq \frac{1}{K}\sum_{i=1}^{K} p(\mathbf{x}|\mathbf{z}_i,\mathbf{w})$$

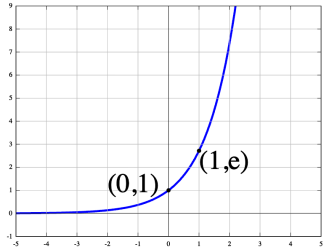


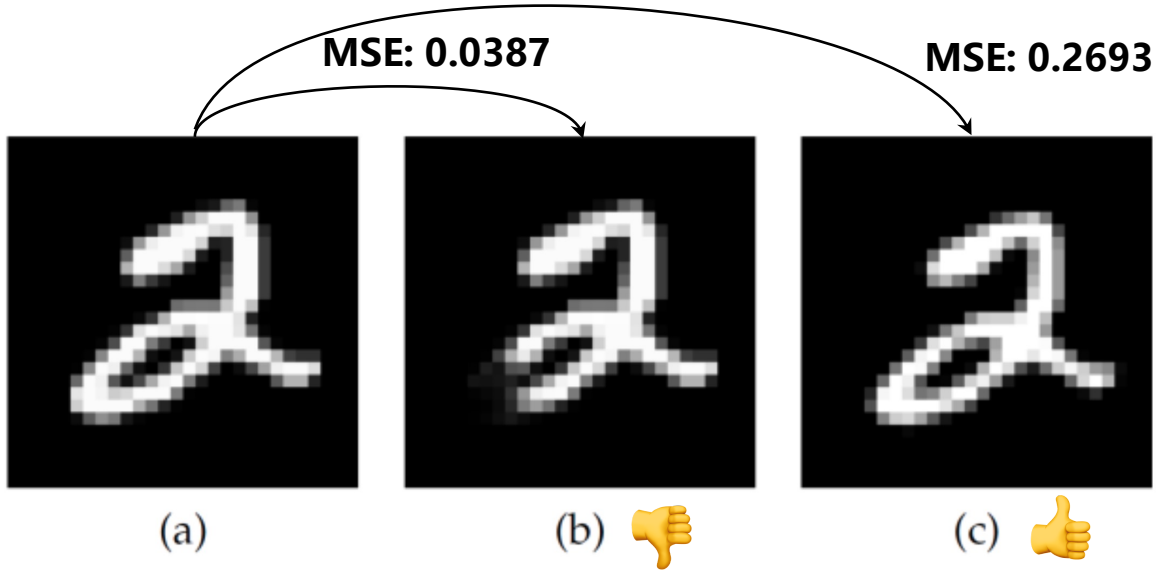


从简单分布中采样

- 训练目标：调整模型的参数，使得模型生成训练集里真实图片的概率（似然 $p(\mathbf{x})$）尽可能大。

$$p(\mathbf{x}|\mathbf{z}) = C \cdot \exp\left(-\frac{\|\mathbf{x} - \boldsymbol{\mu}_{\mathrm{w}}(\mathbf{z})\|_2^2}{2\sigma^2}\right) = C \cdot \exp\left(-\frac{D \cdot \mathbf{MSE}}{2\sigma^2}\right)$$



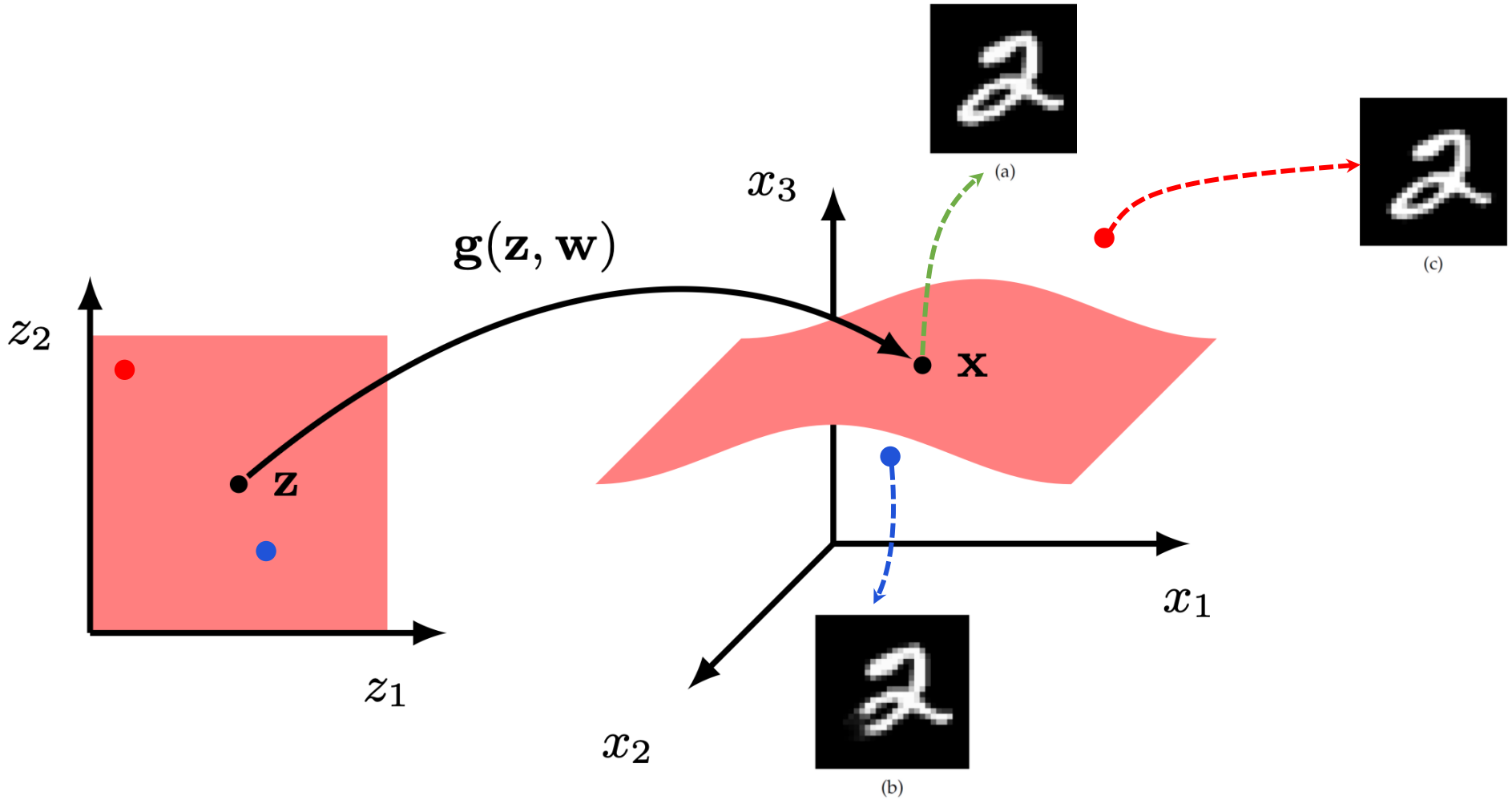


真实样本图像　　生成的损坏的图　　生成的偏移的图

- 对于b：MSE小，似然大。但是本来应该似然小，所以，应该设置σ小；
- 对于c：MSE大，似然小。如果设置了σ小，那么似然会进一步小。

完全随机从z中采用，通过生成的样本与训练样本图像比较似然，这种暴力随机采样法在实际应用中不可行（K的值需要大得离谱）。

从简单分布中采样



本章主要内容

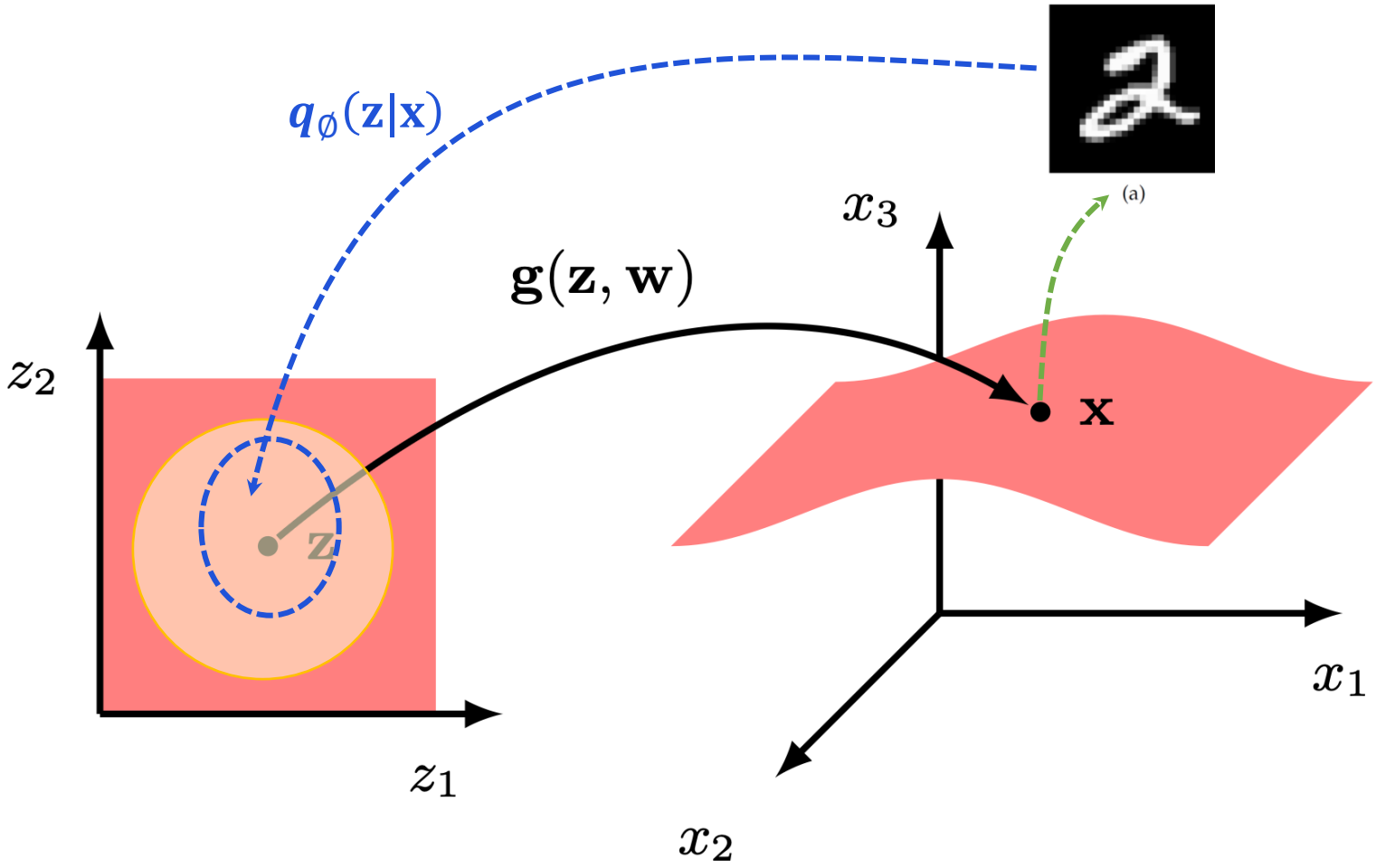
- 生成式模型概述
- 变分自编码器（VAE）
- 扩散模型
- 基于分数的生成

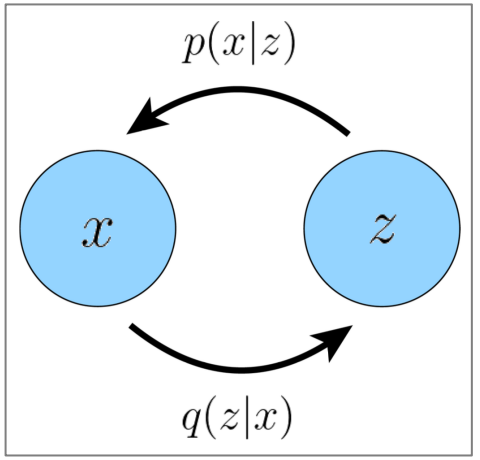
Variational Autoencoder (VAE)

- VAE的思想
  - 放弃盲目从Z中采样，改成先“看一眼”目标，然后划定一个极小的专属片区去采样；
  - 同时放弃计算绝对完美的概率积分，改成优化一个相对容易的数学下界。

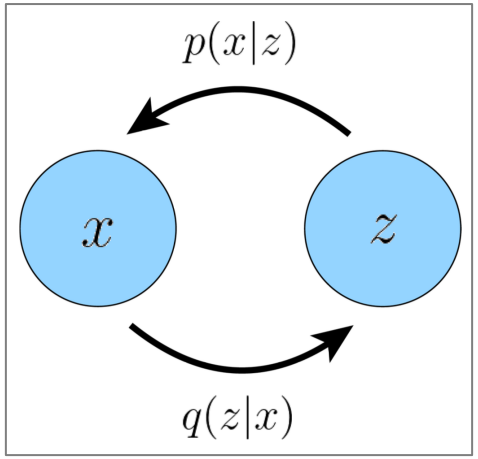
注：本章的模型分布表示为 $p_{\mathrm{w}}(x)$ 或者 $\mathrm{p}(x|\mathrm{w})$

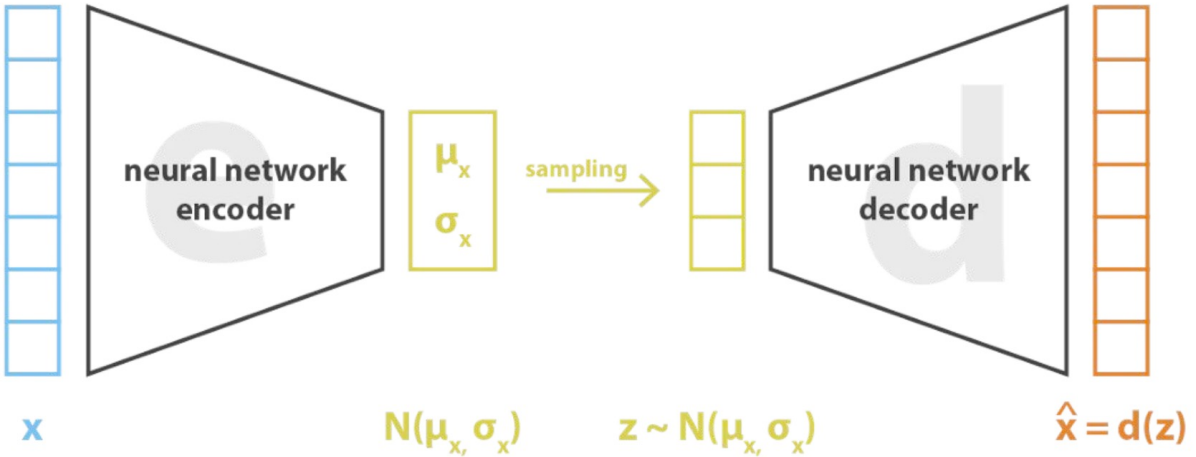
Variational Autoencoder (VAE)



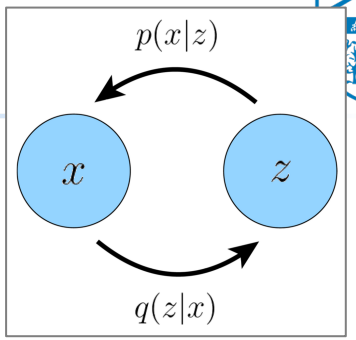


Variational Autoencoder (VAE)

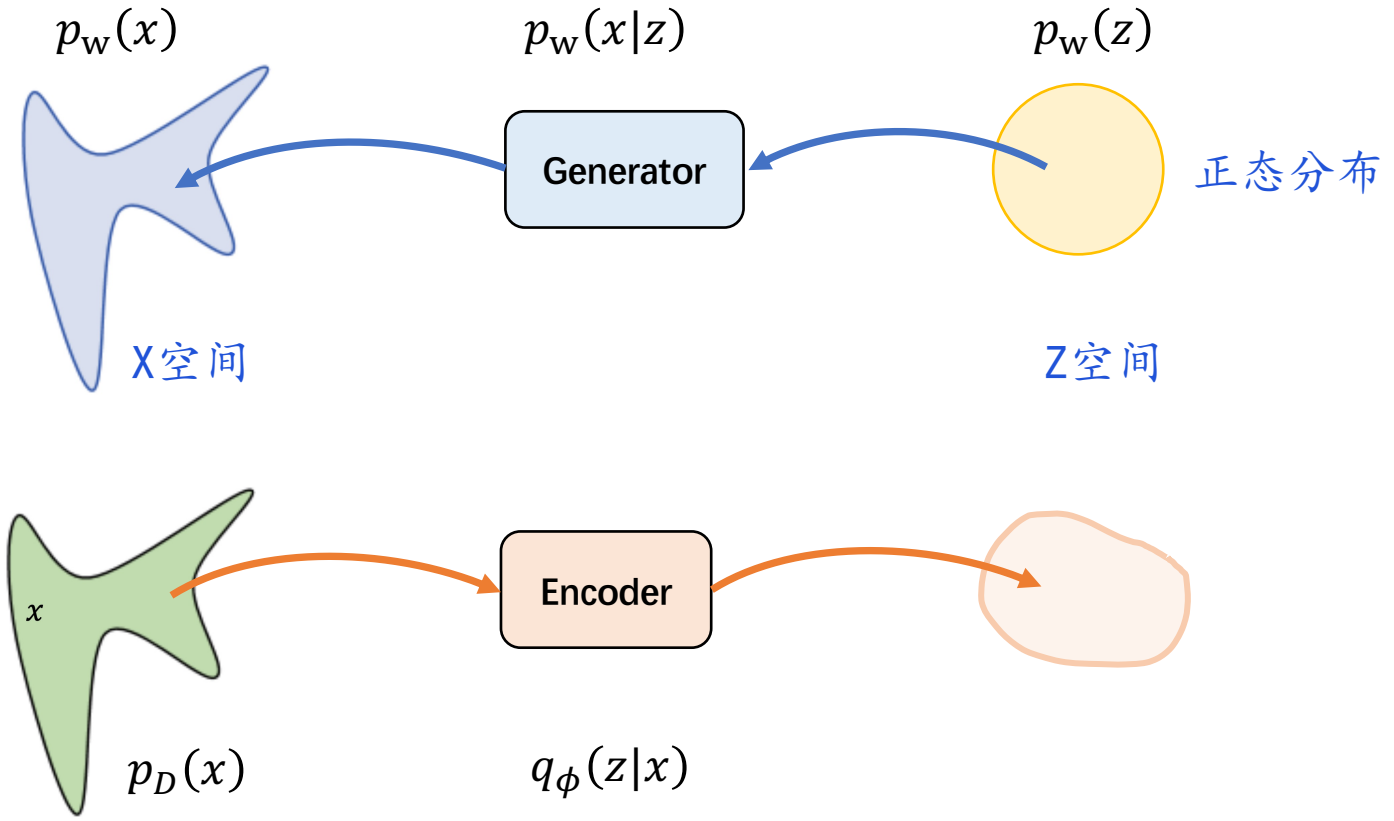




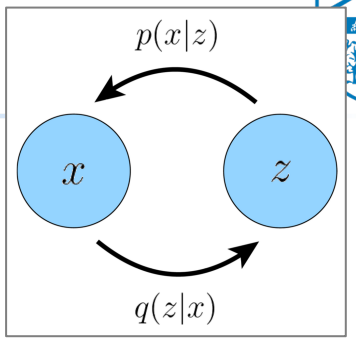
Variational Autoencoder (VAE)



- 从隐变量中生成



Variational Autoencoder (VAE)



- 计算 $\log p_w(x)$

$$\log p_w(x) = \log p_w(x) \int q_\phi(z|x)\, dz$$

$$= \mathbb{E}_{q_\phi(z|x)}[\log p_w(x)] = \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_w(x,z) \cdot q_\phi(z|x)}{p_w(z|x) \cdot q_\phi(z|x)}\right]$$

$$= \underbrace{\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_w(x,z)}{q_\phi(z|x)}\right]}_{\text{ELBO}} + \underbrace{\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{q_\phi(z|x)}{p_w(z|x)}\right]}_{D_{KL}\left(q_\phi(z|x) \,\|\, p_w(z|x)\right)}$$

后验概率$p_w(z|x)$未知，无法计算

$$\geq \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_w(x,z)}{q_\phi(z|x)}\right] = \mathcal{L}(w, \emptyset)$$

证据下界 Evidence Lower Bound (ELBO)

$$\log p_w(x) = \mathcal{L}(w, \emptyset) + D_{KL}\left(q_\phi(z|x) \,\|\, p_w(z|x)\right)$$

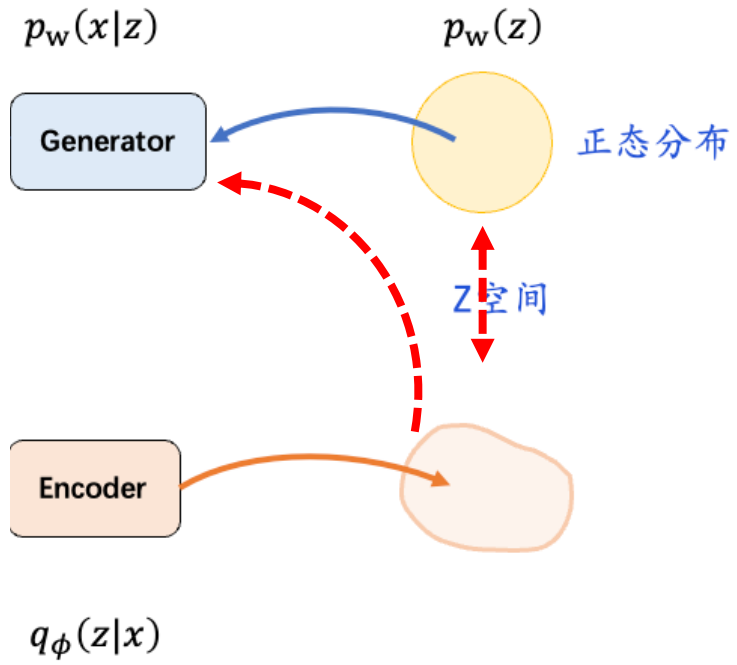
VAE

- **ELBO分解**

$$\mathcal{L}(\mathrm{w},\emptyset) = \mathbb{E}_{q_\phi(z|x)}\left[\log\frac{p_\mathrm{w}(x,z)}{q_\phi(z|x)}\right] = \mathbb{E}_{q_\phi(z|x)}\left[\log\frac{p_\mathrm{w}(z)p_\mathrm{w}(x|z)}{q_\phi(z|x)}\right]$$

$$= \mathbb{E}_{q_\phi(z|x)}\left[\log p_\mathrm{w}(x|z)\right] + \mathbb{E}_{q_\phi(z|x)}\left[\log\frac{p_\mathrm{w}(z)}{q_\phi(z|x)}\right]$$

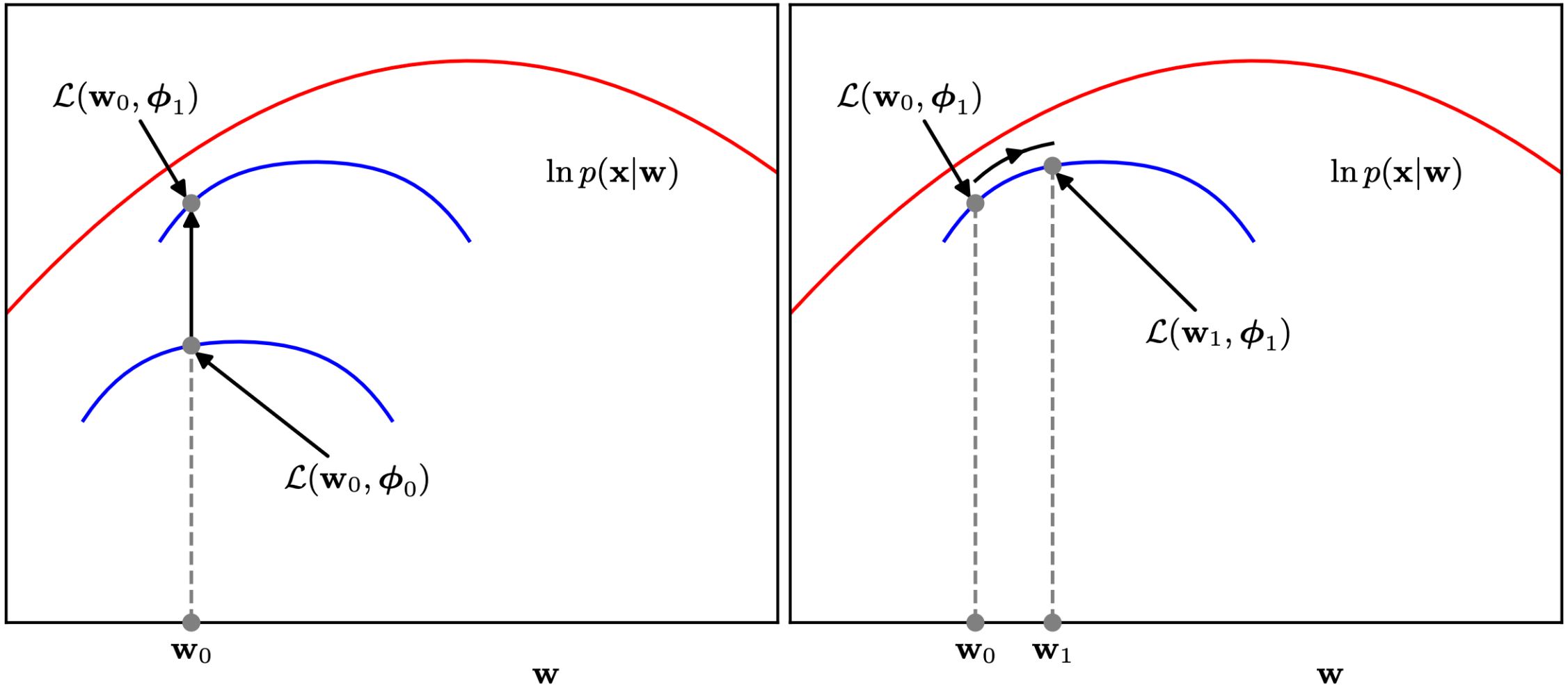
$$= \underbrace{\mathbb{E}_{q_\phi(z|x)}\left[\log p_\mathrm{w}(x|z)\right]}_{\text{重构}} - \underbrace{D_{KL}\left(q_\phi(z|x)\,\|\,p_\mathrm{w}(z)\right)}_{\text{先验匹配}}$$



$$\max \mathbb{E}_{\boldsymbol{p_D(x)}}\left[\mathbb{E}_{q_\phi(z|x)}[\log p_\mathrm{w}(x|z)] - D_{KL}\left(q_\phi(z|x)\,\|\,p_\mathrm{w}(z)\right)\right]$$

VAE

$$\max \mathbb{E}_{p_D(x)} \left[\mathbb{E}_{q_\phi(z|x)} [\log p_w(x|z)] - D_{KL} \left(q_\phi(z|x) \parallel p_w(z) \right) \right]$$



固定w，优化∅

固定∅，优化w

VAE

$$\max \mathbb{E}_{p_D(x)}\left[\mathbb{E}_{q_\phi(z|x)}[\log p_w(x|z)] - D_{KL}\left(q_\phi(z|x) \,\|\, p_w(z)\right)\right]$$

$$\mathbb{E}_{p_D(x)}\left[\mathbb{E}_{q_\phi(z|x)}[\log p_w(x|z)] + \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_w(z)}{q_\phi(z|x)}\right]\right]$$

$$= \mathbb{E}_{q_{D,\phi}(x,z)}[\log p_w(x|z) + \log p_w(z) - \log q_\phi(z|x)]$$

$$= \mathbb{E}_{q_{D,\phi}(x,z)}\left[\log \frac{p_w(x,z)}{q_\phi(z|x)}\right]$$

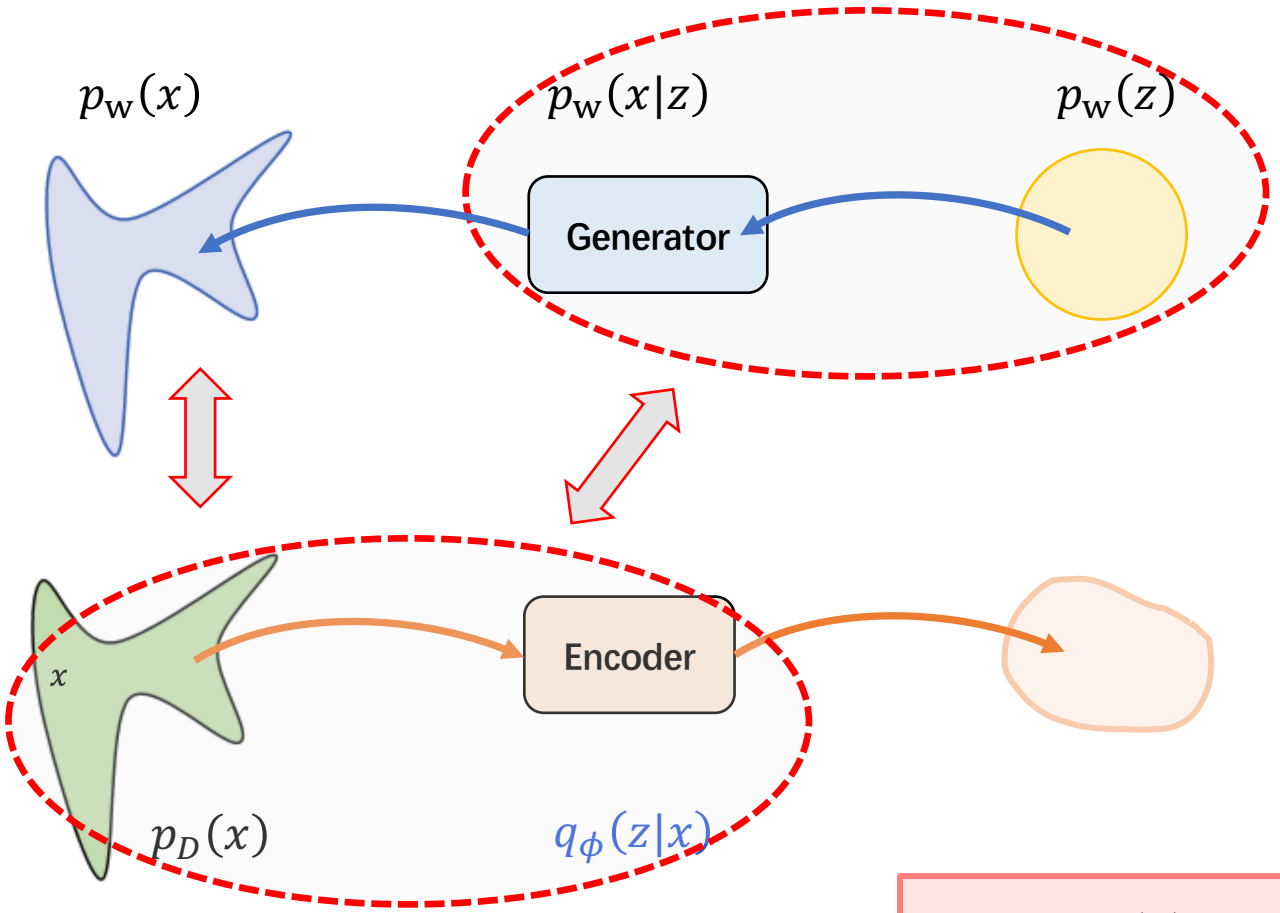
联合推断分布:
$q_{D,\phi}(x,z) = p_D(x)\, q_\phi(z|x)$

$$= \mathbb{E}_{q_{D,\phi}(x,z)}\left[\log \frac{p_w(x,z)}{q_\phi(x,z)} + \log p_D(x)\right]$$

$$= -\underbrace{D_{KL}\left(q_\phi(x,z) \,\|\, p_w(x,z)\right)}_{\text{联合概率之间的距离}} + \underbrace{\mathbb{E}_{p_D(x)}[\log p_D(x)]}_{\text{常量}}$$

VAE

$$D_{KL}\left(p_D(x) \parallel p_w(x)\right) \Rightarrow D_{KL}\left(q_\phi(x,z) \parallel p_w(x,z)\right)$$

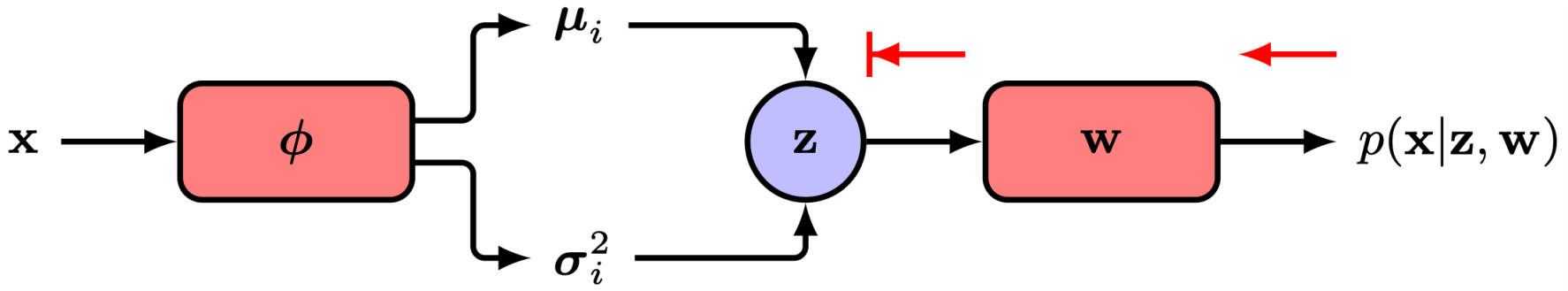


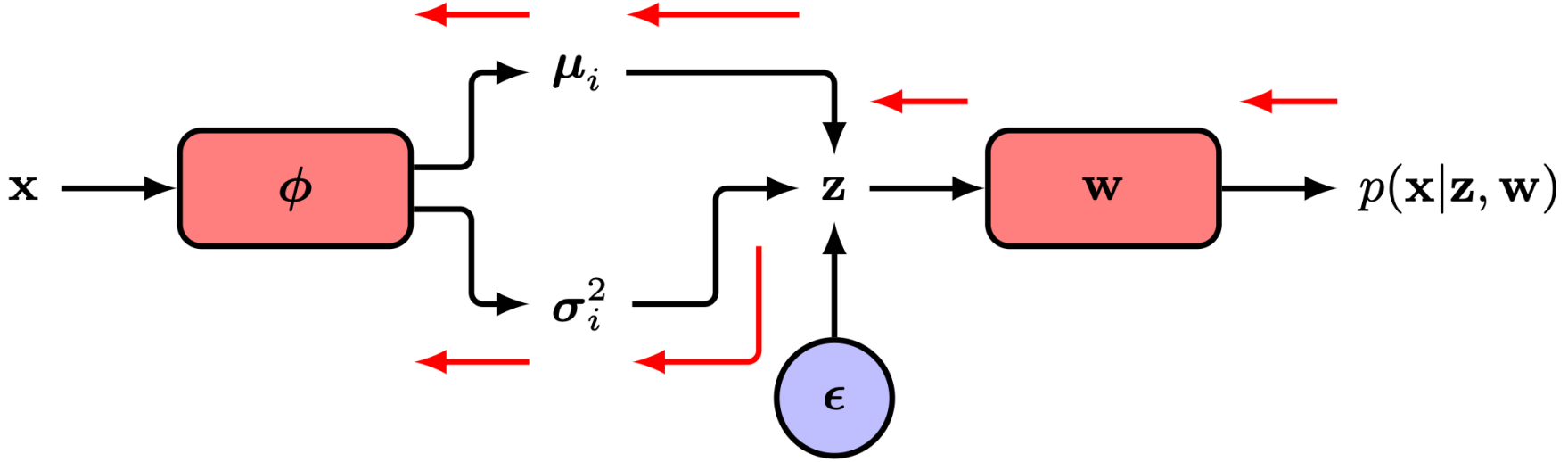
$$\log p_w(x) = \mathcal{L}(w, \emptyset) + D_{KL}\left(q_\phi(z|x) \parallel p_w(z|x)\right)$$

联合概率与边缘概率之间差了一个后验，该后验就是不等式扔掉的项

VAE

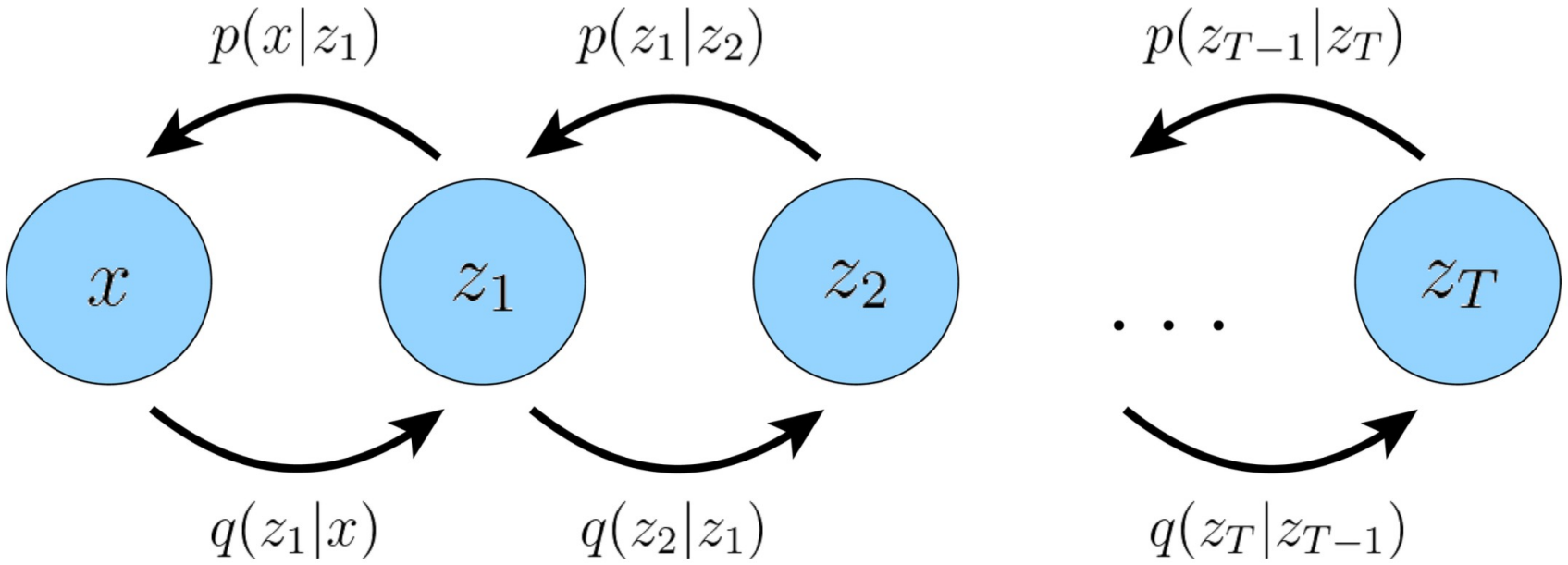
- 重参数化

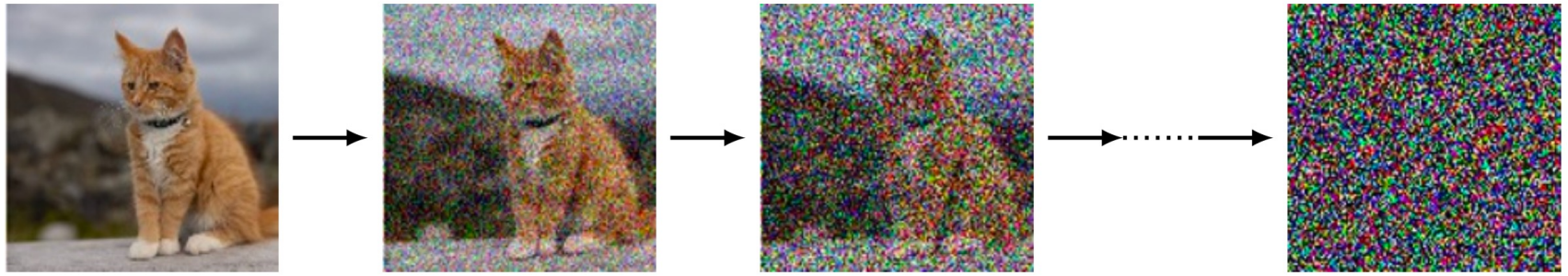

μ_i
$\mathbf{x}$
ϕ
$\mathbf{z}$
$\mathbf{w}$
$p(\mathbf{x}|\mathbf{z},\mathbf{w})$
σ_i^2


μ_i
$\mathbf{x}$
ϕ
$\mathbf{z}$
$\mathbf{w}$
$p(\mathbf{x}|\mathbf{z},\mathbf{w})$
σ_i^2
ϵ

从层级VAE到扩散模型

- 把编码过程变成一个**确定的加噪声过程**





本章主要内容

- 生成式模型概述
- 变分自编码器（VAE）
- 扩散模型
- 基于分数的生成

扩散模型 — 编码过程

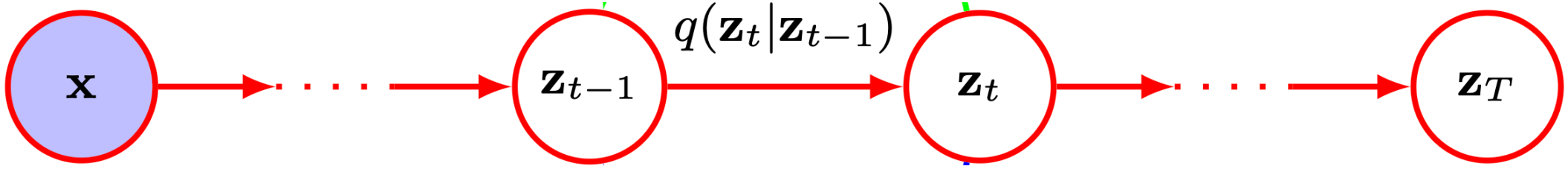
- 把编码过程变成一个**确定的加噪声过程（也称为前向过程）**

$$\mathbf{z}_1 = \sqrt{1-\beta_1}\mathbf{x} + \sqrt{\beta_1}\boldsymbol{\epsilon}_1 \qquad \boldsymbol{\epsilon}_1 \sim \mathcal{N}(\boldsymbol{\epsilon}_1|\mathbf{0},\mathbf{I})$$

β_1为噪声分布的方差

$$q(\mathbf{z}_1|\mathbf{x}) = \mathcal{N}(\mathbf{z}_1|\sqrt{1-\beta_1}\mathbf{x}, \beta_1\mathbf{I}) \qquad q(\mathbf{z}_t|\mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t|\sqrt{1-\beta_t}\mathbf{z}_{t-1}, \beta_t\mathbf{I})$$

该条件分布也为高斯分布



扩散模型 — 编码过程

- 在观测数据x的条件下，隐变量的联合分布

$$q(\mathbf{z}_1,\ldots,\mathbf{z}_t|\mathbf{x}) = q(\mathbf{z}_1|\mathbf{x})\prod_{\tau=2}^{t} q(\mathbf{z}_\tau|\mathbf{z}_{\tau-1})$$

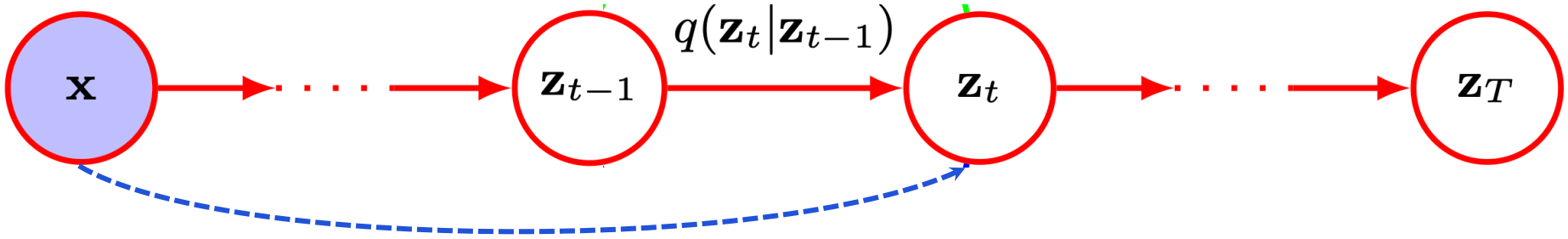
- 扩散核（ Diffusion kernel ）

$$q(\mathbf{z}_t|\mathbf{x}) = \mathcal{N}(\mathbf{z}_t|\sqrt{\alpha_t}\mathbf{x},(1-\alpha_t)\mathbf{I}) \qquad \alpha_t = \prod_{\tau=1}^{t}(1-\beta_\tau)$$

$$\mathbf{z}_t = \sqrt{\alpha_t}\mathbf{x} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t$$

$$q(\mathbf{z}_T|\mathbf{x}) = \mathcal{N}(\mathbf{z}_T|\mathbf{0},\mathbf{I})$$

完全的标准正态分布



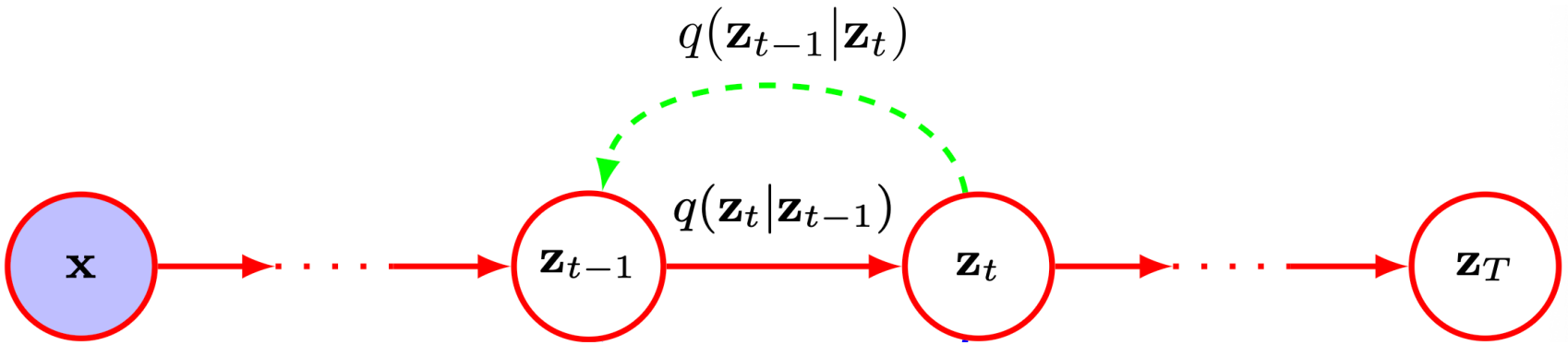
扩散模型 — 编码过程

- 我们目标是逆转加噪过程，考虑条件概率分布 $q(z_{t-1}|z_t)$

$$q(\mathbf{z}_{t-1}|\mathbf{z}_t) = \frac{q(\mathbf{z}_t|\mathbf{z}_{t-1})q(\mathbf{z}_{t-1})}{q(\mathbf{z}_t)}$$

$$q(\mathbf{z}_{t-1}) = \int q(\mathbf{z}_{t-1}|\mathbf{x})p(\mathbf{x})\,\mathrm{d}\mathbf{x}$$

- 需要针对p(x)进行积分，P(x)正是我们要建模的分布。因此，计算$q(z_{t-1})$很困难
- 如果通过采样来近似，会得到以高斯混合形式表示的复杂分布



扩散模型 — 编码过程

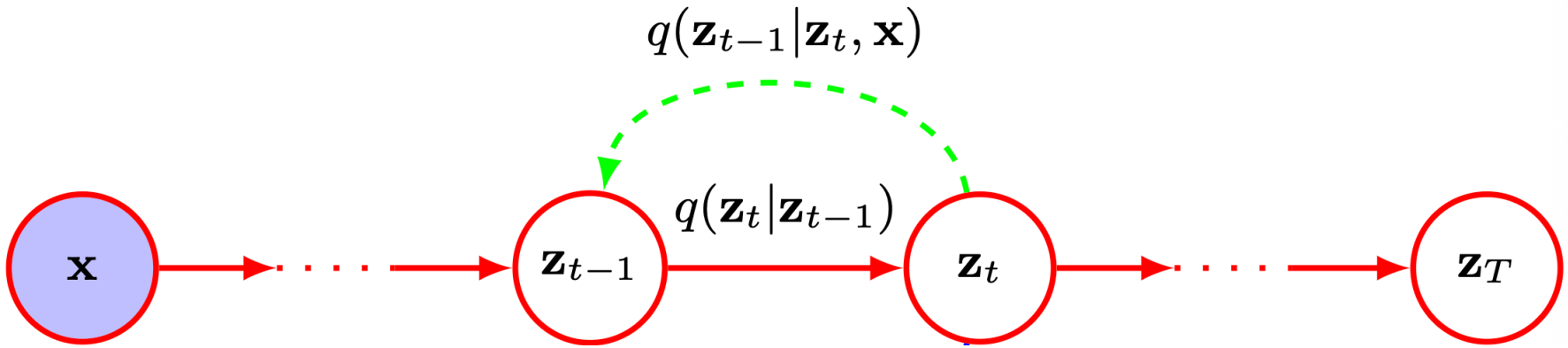
- 考虑逆向分布的条件版本，以x为条件的概率分布 $\mathbf{q(z_{t-1}|z_t, x)}$

$$q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x}) = q(\mathbf{z}_t|\mathbf{z}_{t-1})$$

$$q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) = \frac{q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x})q(\mathbf{z}_{t-1}|\mathbf{x})}{q(\mathbf{z}_t|\mathbf{x})}$$

Diffusion kernel

关于z_{t-1}的函数，与z_t无关



扩散模型 — 编码过程

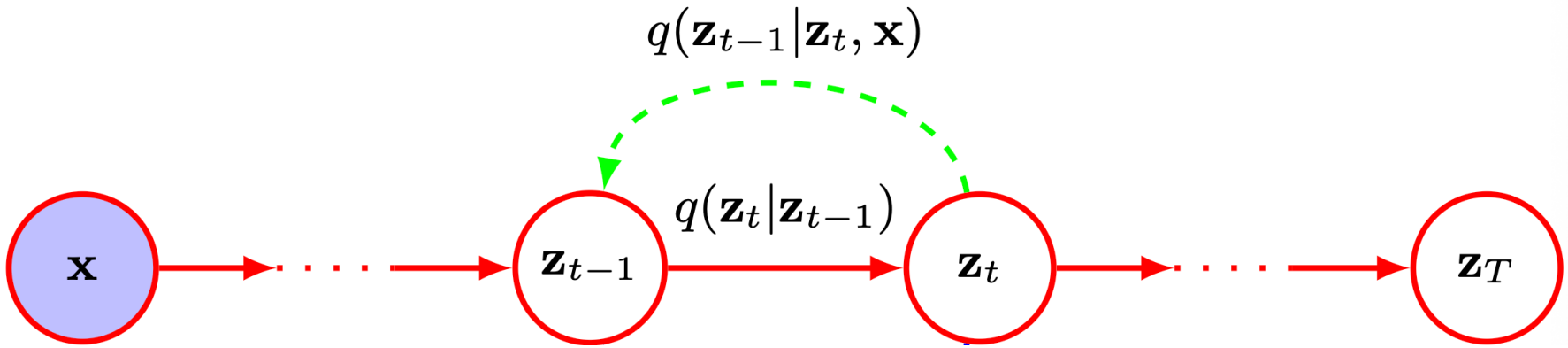
- 考虑逆向分布的条件版本，以x为条件的概率分布 $\mathbf{q(z_{t-1}|z_t, x)}$

$$q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) = \mathcal{N}\left(\mathbf{z}_{t-1}|\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t), \sigma_t^2\mathbf{I}\right)$$

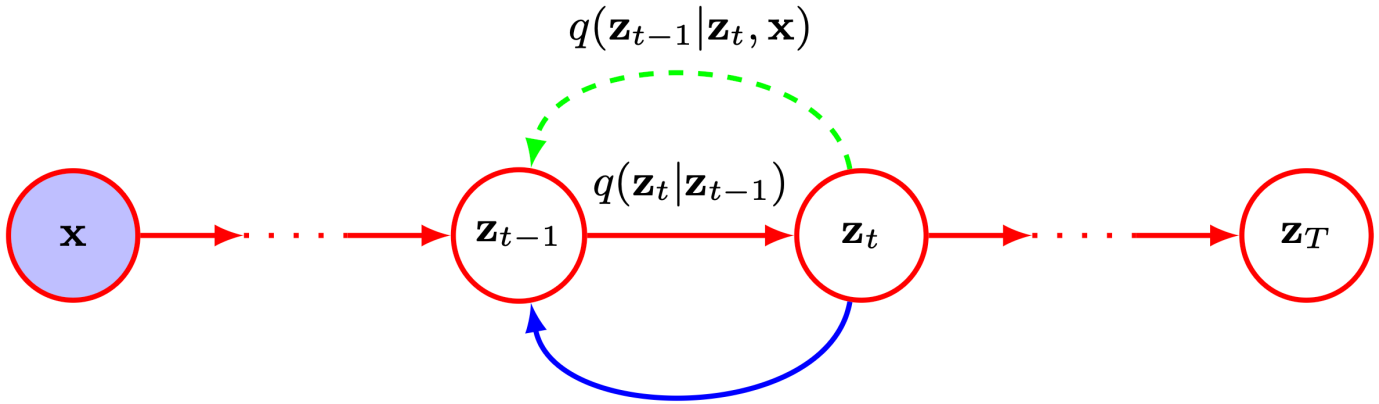
$$\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) = \frac{(1-\alpha_{t-1})\sqrt{1-\beta_t}\mathbf{z}_t + \sqrt{\alpha_{t-1}}\beta_t\mathbf{x}}{1-\alpha_t}$$

用x和$\mathbf{z_t}$来表示均值

$$\sigma_t^2 = \frac{\beta_t(1-\alpha_{t-1})}{1-\alpha_t}$$



扩散模型 — 解码过程

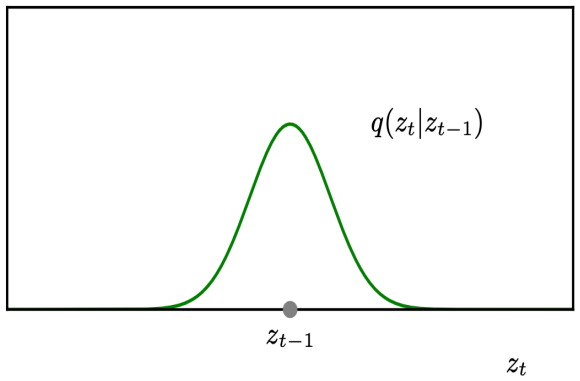


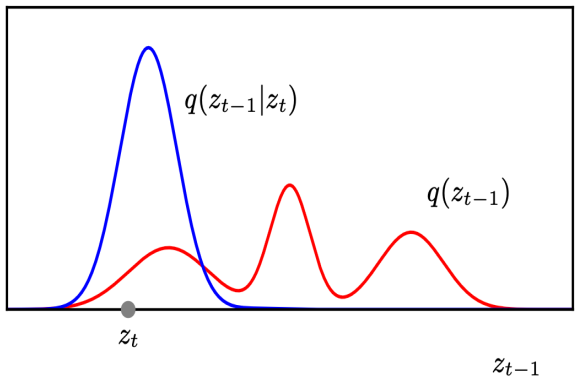
用神经网络来建模 $\mathbf{p}(\mathbf{z_{t-1}}|\mathbf{z}_t, \mathbf{w})$ ，以近似 $\mathbf{q}(\mathbf{z_{t-1}}|\mathbf{z}_t)$

$p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})$

$$q(\mathbf{z}_{t-1}|\mathbf{z}_t) = \frac{q(\mathbf{z}_t|\mathbf{z}_{t-1})q(\mathbf{z}_{t-1})}{q(\mathbf{z}_t)}$$

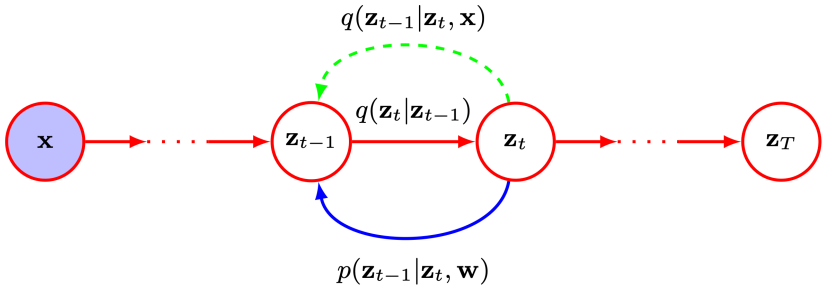
$$p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}) = \mathcal{N}(\mathbf{z}_{t-1}|\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t), \beta_t \mathbf{I})$$





当方差很小时，$q(z_{t-1}|z_t)$也近似为高斯

扩散模型 — 解码过程



- **解码过程的联合概率分布**

$$p(\mathbf{x}, \mathbf{z}_1, \ldots, \mathbf{z}_T|\mathbf{w}) = p(\mathbf{z}_T)\left\{\prod_{t=2}^{T} p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})\right\} p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})$$

- **似然：** $p(\mathbf{x}|\mathbf{w}) = \int \cdots \int p(\mathbf{x}, \mathbf{z}_1, \ldots, \mathbf{z}_T|\mathbf{w})\, \mathrm{d}\mathbf{z}_1 \ldots \mathrm{d}\mathbf{z}_T$

针对高度负责的神经网络函数算积分，很困难

$$\ln p(\mathbf{x}|\mathbf{w}) = \mathcal{L}(\mathbf{w}) + \mathrm{KL}\left(q(\mathbf{z})\|p(\mathbf{z}|\mathbf{x}, \mathbf{w})\right)$$

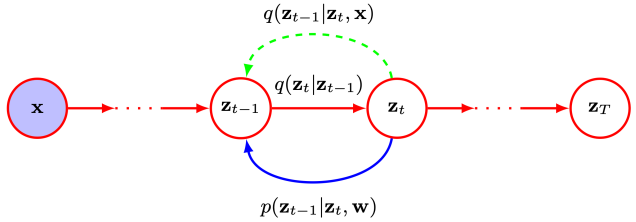
对任意的q分布成立

$$\mathrm{KL}\left(f(\mathbf{z})\|g(\mathbf{z})\right) = -\int f(\mathbf{z}) \ln\left\{\frac{g(\mathbf{z})}{f(\mathbf{z})}\right\} \mathrm{d}\mathbf{z}$$

- **ELBO：** $\mathcal{L}(\mathbf{w}) = \int q(\mathbf{z}) \ln\left\{\frac{p(\mathbf{x}, \mathbf{z}|\mathbf{w})}{q(\mathbf{z})}\right\} \mathrm{d}\mathbf{z}$

近似的优化目标

扩散模型 — 解码过程



- **证据下界ELBO：**

$$\mathcal{L}(\mathbf{w}) = \int q(\mathbf{z}) \ln \left\{ \frac{p(\mathbf{x}, \mathbf{z}|\mathbf{w})}{q(\mathbf{z})} \right\} \mathrm{d}\mathbf{z}$$

$$q(\mathbf{z}_1, \ldots, \mathbf{z}_t|\mathbf{x}) = q(\mathbf{z}_1|\mathbf{x}) \prod_{\tau=2}^{t} q(\mathbf{z}_\tau|\mathbf{z}_{\tau-1})$$

$$p(\mathbf{x}, \mathbf{z}_1, \ldots, \mathbf{z}_T|\mathbf{w}) = p(\mathbf{z}_T) \left\{ \prod_{t=2}^{T} p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}) \right\} p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})$$

$$\mathcal{L}(\mathbf{w}) = \mathbb{E}_q \left[\ln \frac{p(\mathbf{z}_T) \left\{ \prod_{t=2}^{T} p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}) \right\} p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})}{q(\mathbf{z}_1|\mathbf{x}) \prod_{t=2}^{T} q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x})} \right]$$

$$= \mathbb{E}_q \left[\ln p(\mathbf{z}_T) + \sum_{t=2}^{T} \ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x})} - \ln q(\mathbf{z}_1|\mathbf{x}) + \ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w}) \right]$$

和参数无关

需要成对地采样$\mathbf{z_{t-1}}$，$\mathbf{z_t}$

和参数无关

重构项

$$\mathbb{E}_q \left[\ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w}) \right] \simeq \sum_{l=1}^{L} \ln p(\mathbf{x}|\mathbf{z}_1^{(l)}, \mathbf{w})$$

扩散模型 — 解码过程

- **重写ELBO：**

$$\mathcal{L}(\mathbf{w}) = \mathbb{E}_q \left[\ln \frac{p(\mathbf{z}_T) \left\{ \prod_{t=2}^{T} p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}) \right\} p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})}{q(\mathbf{z}_1|\mathbf{x}) \prod_{t=2}^{T} q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x})} \right]$$

$$= \mathbb{E}_q \left[\ln p(\mathbf{z}_T) + \sum_{t=2}^{T} \ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x})} - \ln q(\mathbf{z}_1|\mathbf{x}) + \ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w}) \right]$$

$$q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x}) = \frac{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) q(\mathbf{z}_t|\mathbf{x})}{q(\mathbf{z}_{t-1}|\mathbf{x})}$$

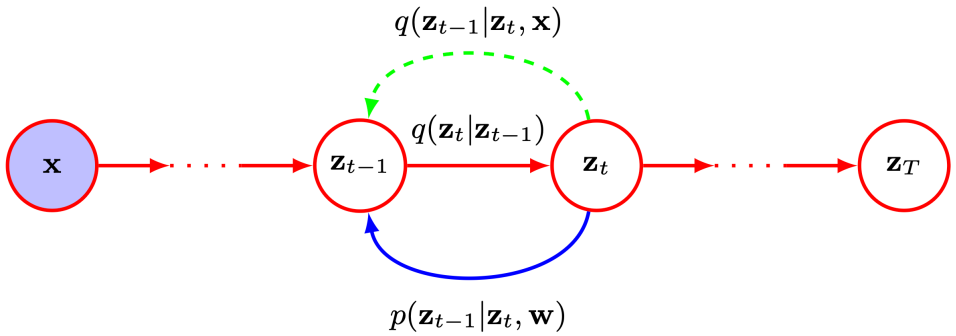
$$\ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x})} = \ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})} + \ln \frac{q(\mathbf{z}_{t-1}|\mathbf{x})}{q(\mathbf{z}_t|\mathbf{x})}$$

$$\mathcal{L}(\mathbf{w}) = \mathbb{E}_q \left[\sum_{t=2}^{T} \ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})} + \ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w}) \right]$$

扩散模型 — 解码过程

$$q(\mathbf{z}_{t-1}, \mathbf{z}_t|\mathbf{x}) = q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})q(\mathbf{z}_t|\mathbf{x})$$

$$\mathcal{L}(\mathbf{w}) = \mathbb{E}_q\left[\sum_{t=2}^{T} \ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})} + \ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})\right]$$



只依赖于 z_t和 z_{t-1}两个变量

$$= \sum_{t=2}^{T} \iint q(\mathbf{z}_{t-1}, \mathbf{z}_t|\mathbf{x}) \ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})} \mathrm{d}\mathbf{z}_{t-1}\mathrm{d}\mathbf{z}_t$$

$$= \sum_{t=2}^{T} \int q(\mathbf{z}_t|\mathbf{x}) \left[\int q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) \ln \frac{p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})}{q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})} \mathrm{d}\mathbf{z}_{t-1}\right] \mathrm{d}\mathbf{z}_t$$

$$\mathcal{L}(\mathbf{w}) = \underbrace{\int q(\mathbf{z}_1|\mathbf{x}) \ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w}) \, \mathrm{d}\mathbf{z}_1}_{\text{reconstruction term}} - \underbrace{\sum_{t=2}^{T} \int \mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) \| p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})) q(\mathbf{z}_t|\mathbf{x}) \, \mathrm{d}\mathbf{z}_t}_{\text{consistency terms}}$$

扩散模型 — 解码过程

- **两个高斯分布之间的KL散度**

$$D_{KL}(P \parallel Q) = \frac{1}{2}\left[\log\frac{|\boldsymbol{\Sigma_2}|}{|\boldsymbol{\Sigma_1}|} - d + \mathrm{Tr}\left(\boldsymbol{\Sigma_2^{-1}\Sigma_1}\right) + (\boldsymbol{\mu_1} - \boldsymbol{\mu_2})^T\boldsymbol{\Sigma_2^{-1}}(\boldsymbol{\mu_1} - \boldsymbol{\mu_2})\right]$$

若两个分布的方差一样

0

d

$$D_{KL}(P \parallel Q) = \frac{1}{2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T(\beta_t\mathbf{I})^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

$$D_{KL}(P \parallel Q) = \frac{1}{2\beta_t}||\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2||^2$$

在方差固定的高斯分布下，两个概率分布的KL散度，在数学上严格等价于拉近它们均值向量的空间欧氏距离（L2 距离）

扩散模型 — 解码过程

$$\mathcal{L}(\mathbf{w}) = \underbrace{\int q(\mathbf{z}_1|\mathbf{x}) \ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w}) \, \mathrm{d}\mathbf{z}_1}_{\text{reconstruction term}} - \underbrace{\sum_{t=2}^{T} \int \mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) \| p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})) q(\mathbf{z}_t|\mathbf{x}) \, \mathrm{d}\mathbf{z}_t}_{\text{consistency terms}}$$

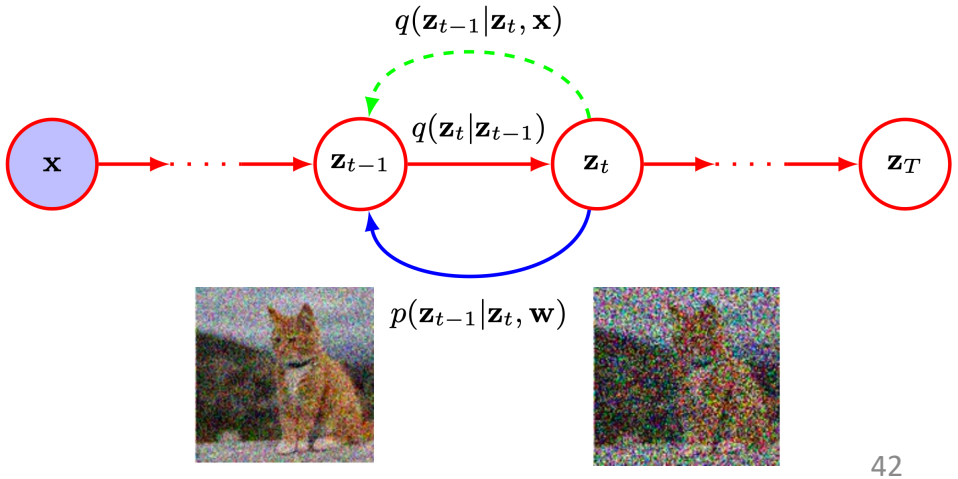
$$q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) = \mathcal{N}\left(\mathbf{z}_{t-1}|\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t), \sigma_t^2 \mathbf{I}\right)$$

$$p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w}) = \mathcal{N}(\mathbf{z}_{t-1}|\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t), \beta_t \mathbf{I})$$

一般地，设定这两个分布的方差一样

$$\mathbb{E}_{\mathbf{z}_t \sim q(\mathbf{z}_t|\mathbf{x})}\left[\frac{1}{2\beta_t}||\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) - \boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t)||^2\right]$$

神经网络预测去噪后的图像



扩散模型 — 解码过程

$$\mathbf{x} = \frac{1}{\sqrt{\alpha_t}}\mathbf{z}_t - \frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}}\boldsymbol{\epsilon}_t$$

$$\mathcal{L}(\mathbf{w}) = \underbrace{\int q(\mathbf{z}_1|\mathbf{x}) \ln p(\mathbf{x}|\mathbf{z}_1, \mathbf{w})\, \mathrm{d}\mathbf{z}_1}_{\text{reconstruction term}} - \underbrace{\sum_{t=2}^{T} \int \mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) \| p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{w})) q(\mathbf{z}_t|\mathbf{x})\, \mathrm{d}\mathbf{z}_t}_{\text{consistency terms}}$$

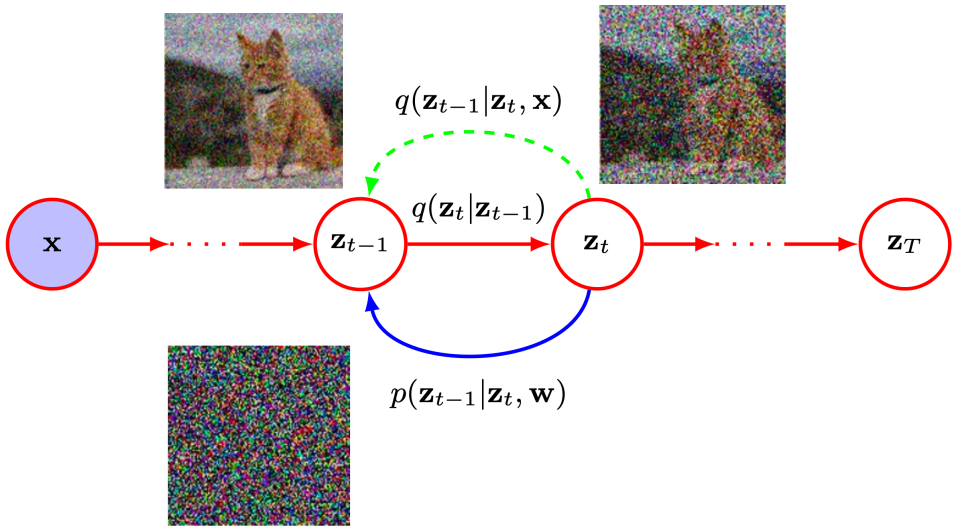
$$\mathbf{m}_t(\mathbf{x}, \mathbf{z}_t) = \frac{1}{\sqrt{1-\beta_t}} \left\{ \mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \boldsymbol{\epsilon}_t \right\}$$

用$\mathbf{z}_t$和噪声$\boldsymbol{\epsilon}_t$来表示均值

$$\boldsymbol{\mu}(\mathbf{z}_t, \mathbf{w}, t) = \frac{1}{\sqrt{1-\beta_t}} \left\{ \mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \mathbf{g}(\mathbf{z}_t, \mathbf{w}, t) \right\}$$

$$\mathcal{L}(\mathbf{w}) = -\sum_{t=1}^{T} \left\| \mathbf{g}(\sqrt{\alpha_t}\mathbf{x} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t, \mathbf{w}, t) - \boldsymbol{\epsilon}_t \right\|^2$$

网络预测噪声



扩散模型 — 解码过程

Algorithm 20.1: Training a denoising diffusion probabilistic model

Input: Training data $\mathcal{D} = \{\mathbf{x}_n\}$

Noise schedule $\{\beta_1, \ldots, \beta_T\}$

Output: Network parameters $\mathbf{w}$

---

for $t \in \{1, \ldots, T\}$ **do**

  $\alpha_t \leftarrow \prod_{\tau=1}^{t}(1-\beta_\tau)$ `// Calculate alphas from betas`

end for

repeat

  $\mathbf{x} \sim \mathcal{D}$ `// Sample a data point`

  $t \sim \{1, \ldots, T\}$ `// Sample a point along the Markov chain`

  $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\epsilon}|\mathbf{0}, \mathbf{I})$ `// Sample a noise vector`

  $\mathbf{z}_t \leftarrow \sqrt{\alpha_t}\mathbf{x} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}$ `// Evaluate noisy latent variable`

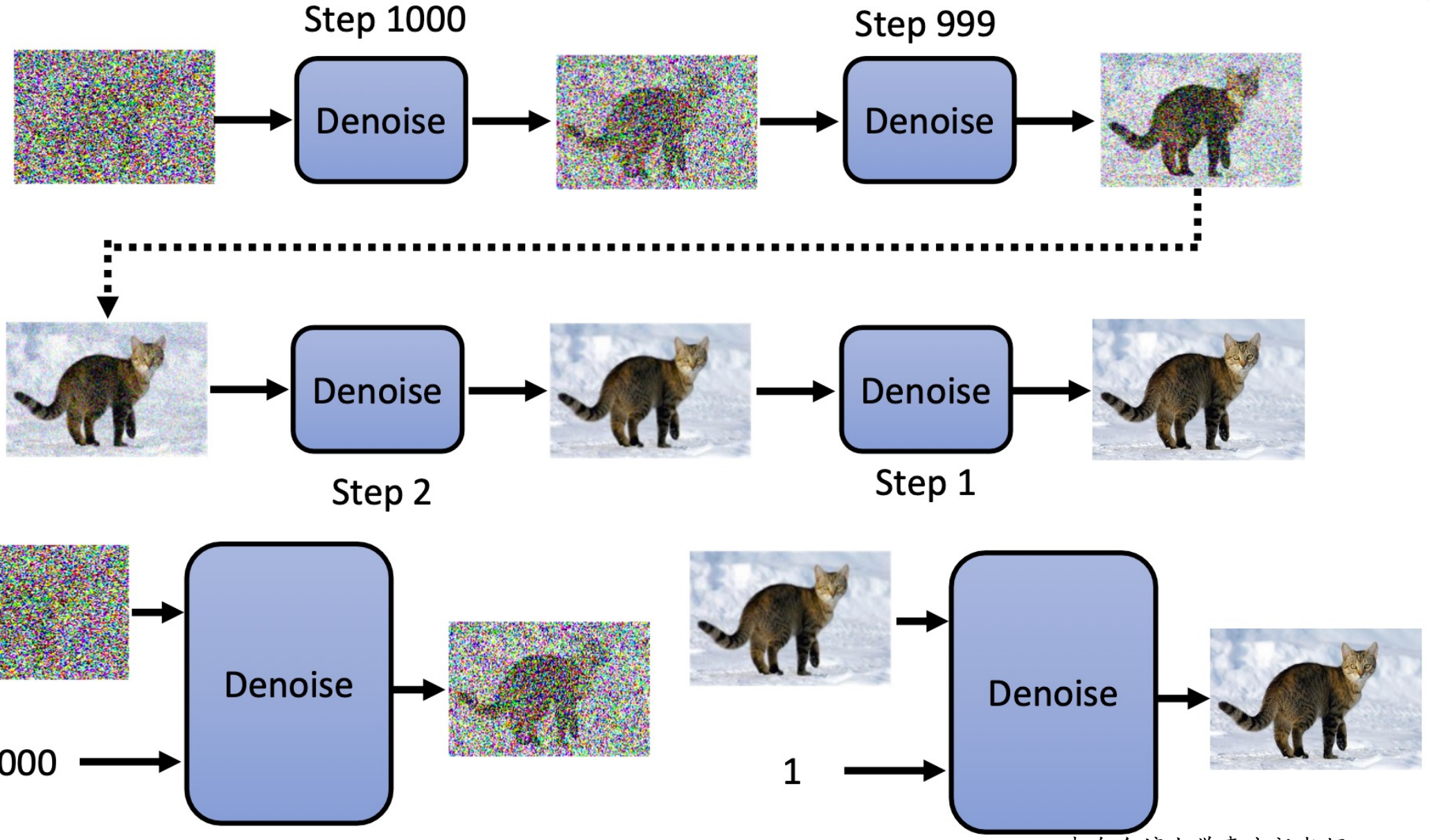
  $\mathcal{L}(\mathbf{w}) \leftarrow \|\mathbf{g}(\mathbf{z}_t, \mathbf{w}, t) - \boldsymbol{\epsilon}\|^2$ `// Compute loss term`

  Take optimizer step

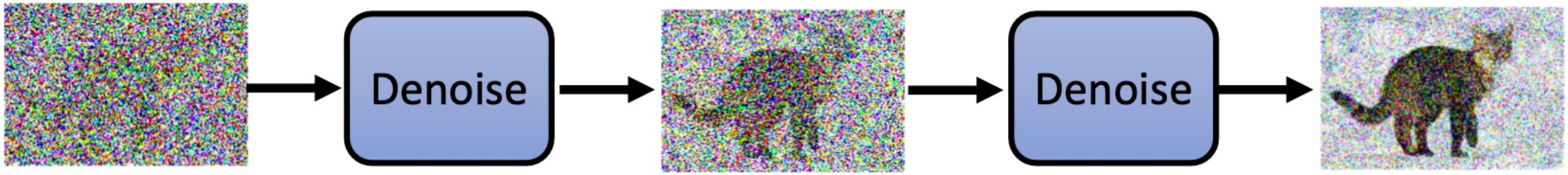
until converged

return $\mathbf{w}$

扩散模型



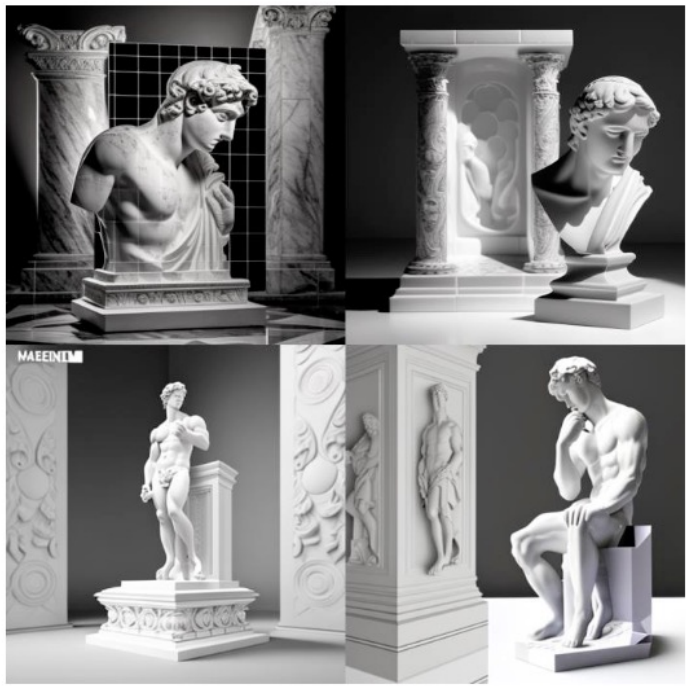
来自台湾大学李宏毅老师



The sculpture is already complete within the marble block, before I start my work. It is already there, I just have to chisel away the superfluous material. - **Michelangelo**

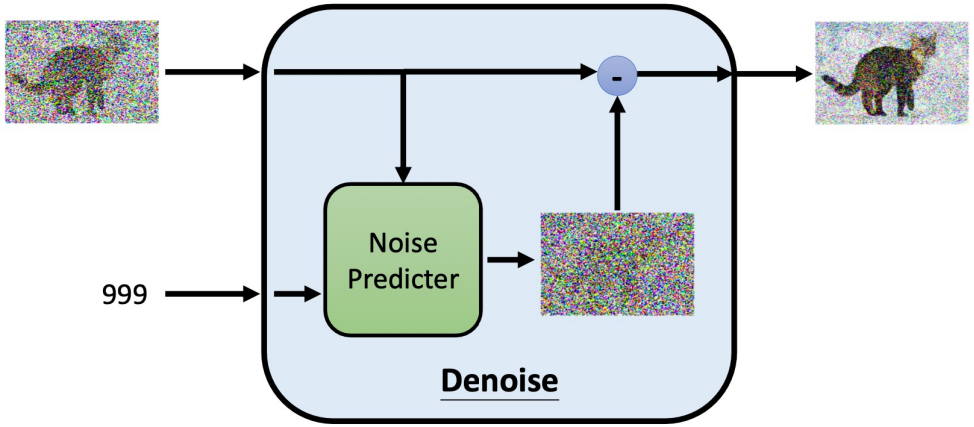


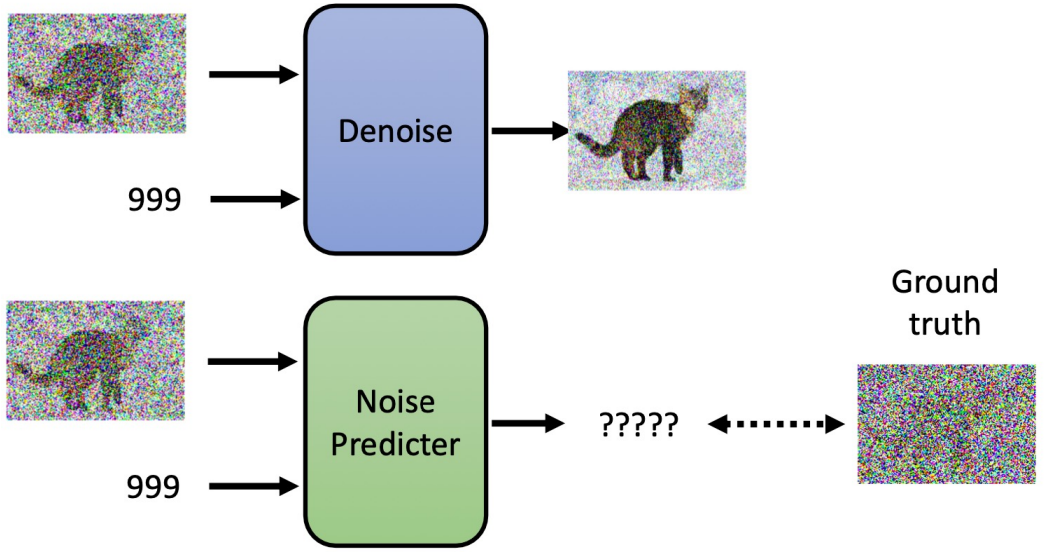


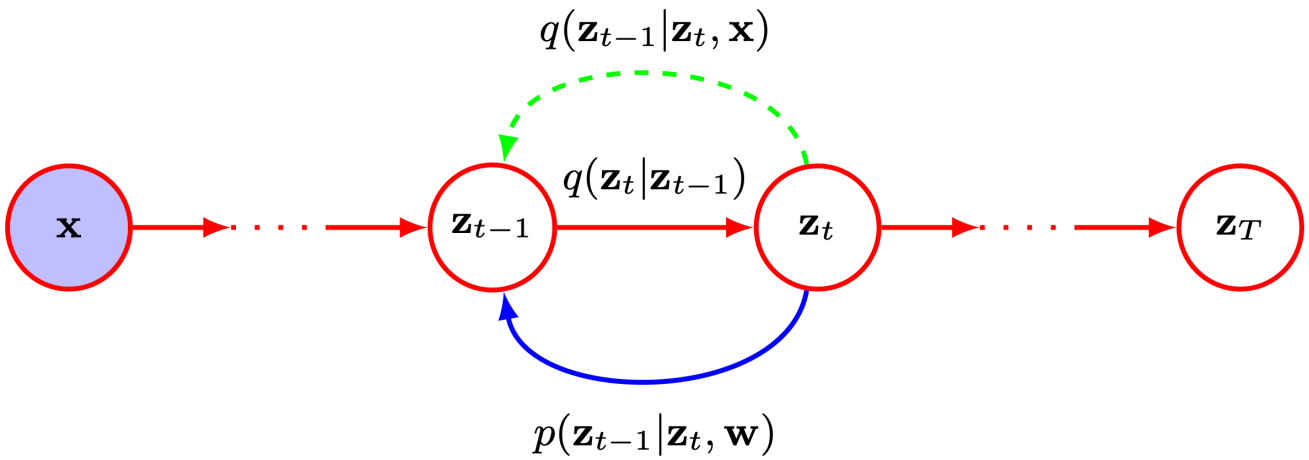


来自台湾大学李宏毅老师

扩散模型







部分内容来自台湾大学李宏毅老师

本章主要内容

- 生成式模型概述
- 变分自编码器（VAE）
- 扩散模型
- 基于分数的生成

能量模型

- 传统的概率模型必须满足 $\int p(\mathbf{x}|\mathbf{w})d\mathbf{x} = 1$ 这个归一化条件。但在极其复杂的高维空间里，想找一个积分等于 1 的复杂函数很困难
- **能量模型：**抛弃归一化条件，定义一个不受限制的实数函数——能量函数 $E(x, w)$，其中w为参数。

概率分布 $$p(\mathbf{x}|\mathbf{w}) = \frac{1}{Z(\mathbf{w})} \exp\{-E(\mathbf{x}, \mathbf{w})\}$$

归一化项 $$Z(\mathbf{w}) = \int \exp\{-E(\mathbf{x}, \mathbf{w})\}\, \mathrm{d}\mathbf{x}$$

ln似然 $$\mathbf{\ln P(x|w) = -E(x, w) - \ln Z(w)}$$

能量模型

$$\ln P(x|w) = -E(x,w) - \ln Z(w)$$

$$\nabla_{\mathbf{w}} \ln Z(\mathbf{w}) = \frac{1}{Z(\mathbf{w})} \nabla_{\mathbf{w}} Z(\mathbf{w}) = \frac{1}{Z(\mathbf{w})} \nabla_{\mathbf{w}} \int \exp(-E(\mathbf{x}, \mathbf{w})) d\mathbf{x}$$

$$Z(\mathbf{w}) = \int \exp\{-E(\mathbf{x}, \mathbf{w})\}\, \mathrm{dx}$$

$$= \frac{1}{Z(\mathbf{w})} \int \nabla_{\mathbf{w}} \exp(-E(\mathbf{x}, \mathbf{w})) d\mathbf{x}$$

$$= \frac{1}{Z(\mathbf{w})} \int [-\exp(-E(\mathbf{x}, \mathbf{w})) \nabla_{\mathbf{w}} E(\mathbf{x}, \mathbf{w})]\, d\mathbf{x}$$

$$-\nabla_{\mathbf{w}} \ln Z(\mathbf{w}) = \int \nabla_{\mathbf{w}} E(\mathbf{x}, \mathbf{w}) \left[\frac{\exp(-E(\mathbf{x}, \mathbf{w}))}{Z(\mathbf{w})} \right] d\mathbf{x}$$

就是概率分布 $\boldsymbol{P(x|w)}$

$$-\nabla_{\mathbf{w}} \ln Z(\mathbf{w}) = \int \{\nabla_{\mathbf{w}} E(\mathbf{x}, \mathbf{w})\} p(\mathbf{x}|\mathbf{w}) d\mathbf{x} = \mathbb{E}_{\mathbf{x} \sim \mathcal{M}} [\nabla_{\mathbf{w}} E(\mathbf{x}, \mathbf{w})]$$

能量函数的梯度相对**模型分布**的期望

从模型中采样

能量模型

$$-\nabla_{\mathbf{w}} \ln Z(\mathbf{w}) = \int \{\nabla_{\mathbf{w}} E(\mathbf{x}, \mathbf{w})\} p(\mathbf{x}|\mathbf{w}) d\mathbf{x} = \mathbb{E}_{\mathbf{x}\sim\mathcal{M}} [\nabla_{\mathbf{w}} E(\mathbf{x}, \mathbf{w})]$$

- 似然函数的梯度

$$\nabla_{\mathbf{w}} \ln p(\mathbf{x}|\mathbf{w}) = -\nabla_{\mathbf{w}} E(\mathbf{x}, \mathbf{w}) - \nabla_{\mathbf{w}} \ln Z(\mathbf{w})$$

$$\mathbb{E}_{\mathbf{x}\sim p_{\mathcal{D}}} [\nabla_{\mathbf{w}} \ln p(\mathbf{x}|\mathbf{w})] = -\mathbb{E}_{\mathbf{x}\sim p_{\mathcal{D}}} [\nabla_{\mathbf{w}} E(\mathbf{x}, \mathbf{w})] - \nabla_{\mathbf{w}} \ln Z(\mathbf{w})$$

$$\nabla_{\mathbf{w}} \mathbb{E}_{\mathbf{x}\sim p_{\mathcal{D}}} [\ln p(\mathbf{x}|\mathbf{w})] = -\mathbb{E}_{\mathbf{x}\sim p_{\mathcal{D}}} [\nabla_{\mathbf{w}} E(\mathbf{x}, \mathbf{w})] + \mathbb{E}_{\mathbf{x}\sim p_{\mathcal{M}}(\mathbf{x})} [\nabla_{\mathbf{w}} E(\mathbf{x}, \mathbf{w})]$$

朝似然梯度方向对w进行微小调整

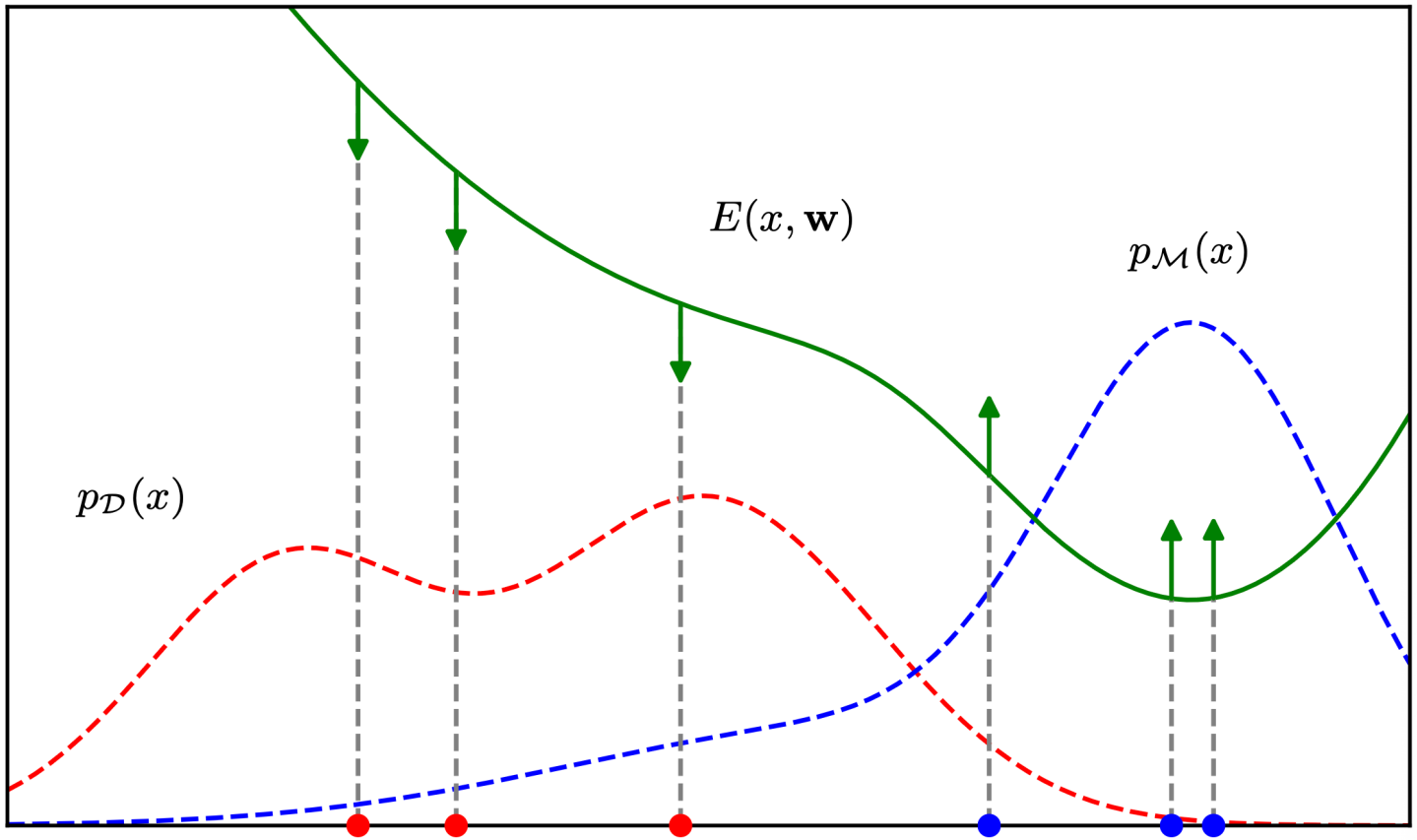
从p_D中采样的样本
$E(x, w)$减小，概率增大

从p_M中采样的样本
$E(x, w)$增大，概率减小

当$p_D = p_M$，上述公式的值为0。

能量模型

$$\nabla_{\mathbf{w}}\mathbb{E}_{\mathbf{x}\sim p_{\mathcal{D}}}\left[\ln p(\mathbf{x}|\mathbf{w})\right] = -\mathbb{E}_{\mathbf{x}\sim p_{\mathcal{D}}}\left[\nabla_{\mathbf{w}}E(\mathbf{x},\mathbf{w})\right] + \mathbb{E}_{\mathbf{x}\sim p_{\mathcal{M}}(\mathbf{x})}\left[\nabla_{\mathbf{w}}E(\mathbf{x},\mathbf{w})\right]$$



对于从p_D中采样的样本本，$E(x,w)$减小，概率增大

对于从p_M中采样的样本，$E(x,w)$增大，概率减小

郎之万动力学（Langevin dynamics）

$$\nabla_{\mathbf{w}}\mathbb{E}_{\mathbf{x}\sim p_{\mathcal{D}}}\left[\ln p(\mathbf{x}|\mathbf{w})\right] = -\mathbb{E}_{\mathbf{x}\sim p_{\mathcal{D}}}\left[\nabla_{\mathbf{w}}E(\mathbf{x},\mathbf{w})\right] + \mathbb{E}_{\mathbf{x}\sim p_{\mathcal{M}}(\mathbf{x})}\left[\nabla_{\mathbf{w}}E(\mathbf{x},\mathbf{w})\right]$$

蒙特卡洛近似

$$\frac{1}{N}\sum_{n=1}^{N}\nabla_{\mathbf{w}}E(\mathbf{x}_n,\mathbf{w})$$

需要从模型分布 $P(x|w)$ 中采样！

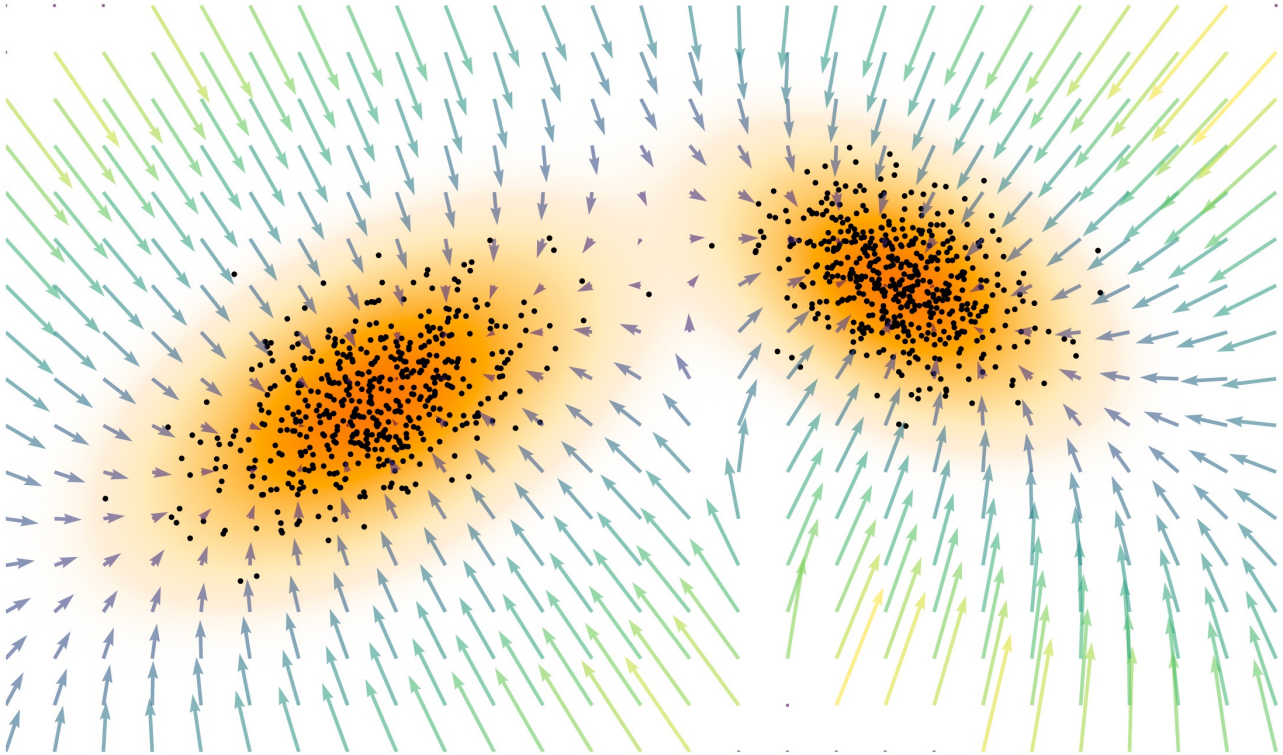
郎之万动力学（Langevin dynamics)

$$p(\mathbf{x}|\mathbf{w}) = \frac{1}{Z(\mathbf{w})} \exp\{-E(\mathbf{x}, \mathbf{w})\}$$

- **得分函数（Score function）**：log似然相对于样本向量x的梯度

$$\mathbf{s}(\mathbf{x}, \mathbf{w}) = \nabla_{\mathbf{x}} \ln p(\mathbf{x}|\mathbf{w}) = -\nabla_{\mathbf{x}} E(\mathbf{x}, \mathbf{w})$$

Score vector fields

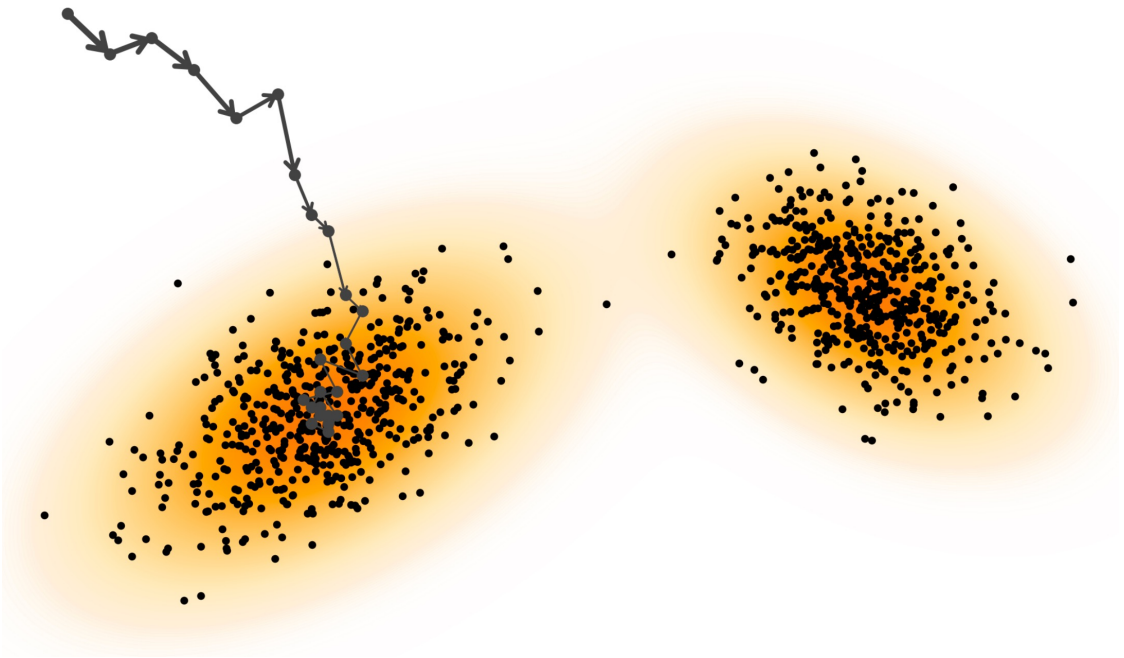


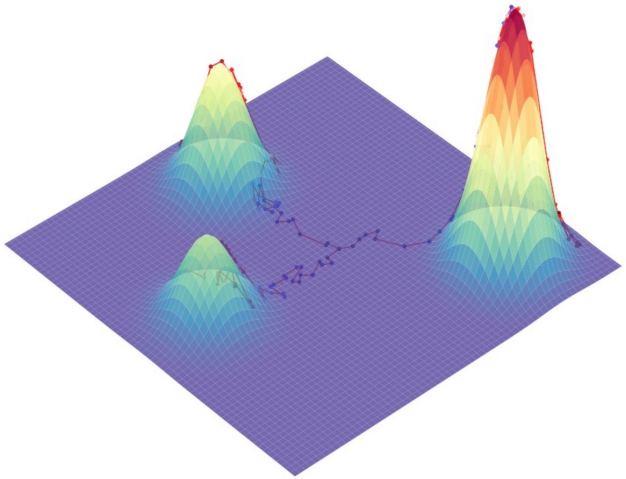
S(x, w)是一个向量值函数，其维度与x相同

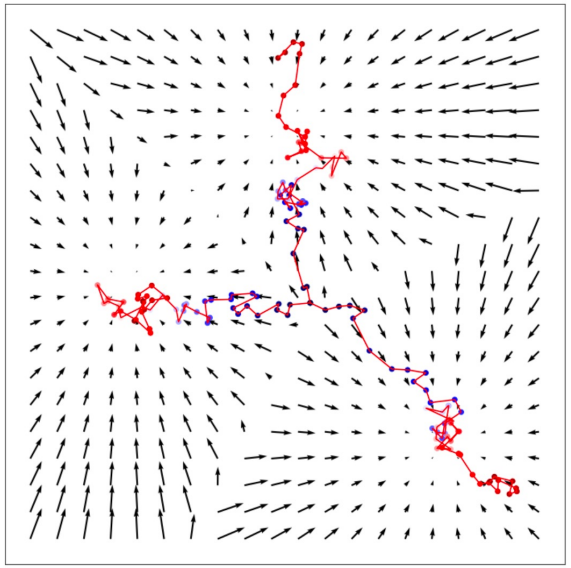
郎之万动力学（Langevin dynamics）

- **郎之万采样**

$$\mathbf{x}^{(\tau+1)} = \mathbf{x}^{(\tau)} + \eta \nabla_{\mathbf{x}} \ln p(\mathbf{x}^{(\tau)}, \mathbf{w}) + \sqrt{2\eta}\boldsymbol{\epsilon}^{(\tau)} \qquad \tau \in 1, \ldots, \mathcal{T}$$







郎之万动力学（Langevin dynamics）

$$\nabla_{\mathbf{w}}\mathbb{E}_{\mathbf{x}\sim p_{\mathcal{D}}}\left[\ln p(\mathbf{x}|\mathbf{w})\right] = -\mathbb{E}_{\mathbf{x}\sim p_{\mathcal{D}}}\left[\nabla_{\mathbf{w}}E(\mathbf{x},\mathbf{w})\right] + \mathbb{E}_{\mathbf{x}\sim p_{\mathcal{M}}(\mathbf{x})}\left[\nabla_{\mathbf{w}}E(\mathbf{x},\mathbf{w})\right]$$

蒙特卡洛近似

$$\frac{1}{N}\sum_{n=1}^{N}\nabla_{\mathbf{w}}E(\mathbf{x}_n,\mathbf{w})$$

基于郎之万采样近似

$$\frac{1}{M}\sum_{m=1}^{M}\nabla_{\mathbf{w}}E(\mathbf{x}_m,\mathbf{w})$$

分数匹配（Score Matching）

- To train such a model we need to define a loss function that aims to match the model score function $\mathrm{s}(\mathrm{x}, \mathrm{w})$ to the score function $\nabla_{\mathrm{x}} \ln \mathrm{p}(\mathrm{x})$ of the distribution $p(x)$ that generated the data.

$$J(\mathbf{w}) = \frac{1}{2} \int \|\mathbf{s}(\mathbf{x}, \mathbf{w}) - \nabla_{\mathbf{x}} \ln p(\mathbf{x})\|^2 p(\mathbf{x}) \, \mathrm{d}\mathbf{x}$$

true data score,
It is unknown

分数匹配（Score Matching）

- 经验分布

$$p_{\mathcal{D}}(\mathbf{x}) = \frac{1}{N}\sum_{n=1}^{N}\delta(\mathbf{x}-\mathbf{x}_n)$$

$$\delta(\mathbf{x}) = 0, \quad \mathbf{x} \neq \mathbf{0}$$

$$\int \delta(\mathbf{x})\,\mathrm{d}\mathbf{x} = 1.$$

- 核密度估计

$$q_\sigma(\mathbf{z}) = \int q(\mathbf{z}|\mathbf{x},\sigma)p(\mathbf{x})\,\mathrm{d}\mathbf{x}$$

- 以每一个训练样本 x为中心，倒扣一个方差为 σ^2的高斯钟形曲线
- 平滑与叠加
- 用$q_\sigma(z)$ 代替P(x)

噪声核（Noise Kernel） $q(\mathbf{z}|\mathbf{x},\sigma) = \mathcal{N}(\mathbf{z}|\mathbf{x},\sigma^2\mathbf{I})$

分数匹配（Score Matching）

$$q_\sigma(\mathbf{z}) = \int q(\mathbf{z}|\mathbf{x},\sigma)p(\mathbf{x})\,\mathrm{d}\mathbf{x}$$

$$J(\mathbf{w}) = \frac{1}{2}\int \|\mathbf{s}(\mathbf{x},\mathbf{w}) - \nabla_{\mathbf{x}} \ln p(\mathbf{x})\|^2 p(\mathbf{x})\,\mathrm{d}\mathbf{x}$$

$$J(\mathbf{w}) = \frac{1}{2}\int \|\mathbf{s}(\mathbf{z},\mathbf{w}) - \nabla_{\mathbf{z}} \ln q_\sigma(\mathbf{z})\|^2 q_\sigma(\mathbf{z})\,\mathrm{d}\mathbf{z}$$

$$J(\mathbf{w}) = \frac{1}{2}\iint \|\mathbf{s}(\mathbf{z},\mathbf{w}) - \nabla_{\mathbf{z}} \ln q(\mathbf{z}|\mathbf{x},\sigma)\|^2 q(\mathbf{z}|\mathbf{x},\sigma)p(\mathbf{x})\,\mathrm{d}\mathbf{z}\,\mathrm{d}\mathbf{x} + \text{const.}$$

$$\nabla_{\mathbf{z}} \ln q(\mathbf{z}|\mathbf{x},\sigma) = -\frac{1}{\sigma}\boldsymbol{\epsilon}$$

去噪得分匹配

补充：高斯分布的梯度

$$q(\mathbf{z}|\mathbf{x},\sigma) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{z}-\mathbf{x}\|^2\right)$$

$$\ln q(\mathbf{z}|\mathbf{x},\sigma) = \ln\left(\frac{1}{(2\pi\sigma^2)^{D/2}}\right) - \frac{1}{2\sigma^2}\|\mathbf{z}-\mathbf{x}\|^2$$

$$\ln q(\mathbf{z}|\mathbf{x},\sigma) = C - \frac{1}{2\sigma^2}\|\mathbf{z}-\mathbf{x}\|^2$$

$$\nabla_{\mathbf{z}} \ln q(\mathbf{z}|\mathbf{x},\sigma) = -\frac{1}{\sigma^2}(\mathbf{z}-\mathbf{x})$$

$$\mathbf{z}-\mathbf{x} = \sigma\boldsymbol{\epsilon}$$

$$\nabla_{\mathbf{z}} \ln q(\mathbf{z}|\mathbf{x},\sigma) = -\frac{1}{\sigma}\boldsymbol{\epsilon}$$

随机微分方差（Stochastic Differential Equation, SDE）

- DDPM中的加噪过程（离散）：

$$\mathbf{z}_t = \sqrt{1-\beta_t}\mathbf{z}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t$$

假设 t是一个**连续的时间变量**，把总时间切分成无数个微小的时间步 Δt。此时，原本离散的 β_t就变成了一个随时间变化的连续函数 $\beta(t)$在极短时间 Δt内的累积量 $\beta_t = \beta(t)\Delta t$

$$\mathbf{z}_{t+\Delta t} = \sqrt{1-\beta(t)\Delta t}\mathbf{z}_t + \sqrt{\beta(t)\Delta t}\boldsymbol{\epsilon}$$

$$\sqrt{1-\beta(t)\Delta t} \approx 1 - \frac{1}{2}\beta(t)\Delta t$$

一阶泰勒展开 $\sqrt{1-x} \approx 1 - \frac{1}{2}x$

$$\mathbf{z}_{t+\Delta t} \approx \left(1 - \frac{1}{2}\beta(t)\Delta t\right)\mathbf{z}_t + \sqrt{\beta(t)}\sqrt{\Delta t}\boldsymbol{\epsilon}$$

$$\mathbf{z}_{t+\Delta t} - \mathbf{z}_t = -\frac{1}{2}\beta(t)\mathbf{z}_t\Delta t + \sqrt{\beta(t)}\sqrt{\Delta t}\boldsymbol{\epsilon}$$

$$\mathrm{d}\mathbf{z} = -\frac{1}{2}\beta(t)\mathbf{z}\mathrm{d}t + \sqrt{\beta(t)}\mathrm{d}\mathbf{w}$$

维纳过程，
即布朗运动

随机微分方差（SDE）

- **正向SDE**

$$\mathrm{d}\mathbf{z} = -\frac{1}{2}\beta(t)\mathbf{z}\mathrm{d}t + \sqrt{\beta(t)}\mathrm{d}\mathbf{w}$$

漂移项 (drift)　　扩散项 (diffusion)

描述了把一张清晰的图片（数据流形）变成一团纯噪声的过程。

- **反向SDE**

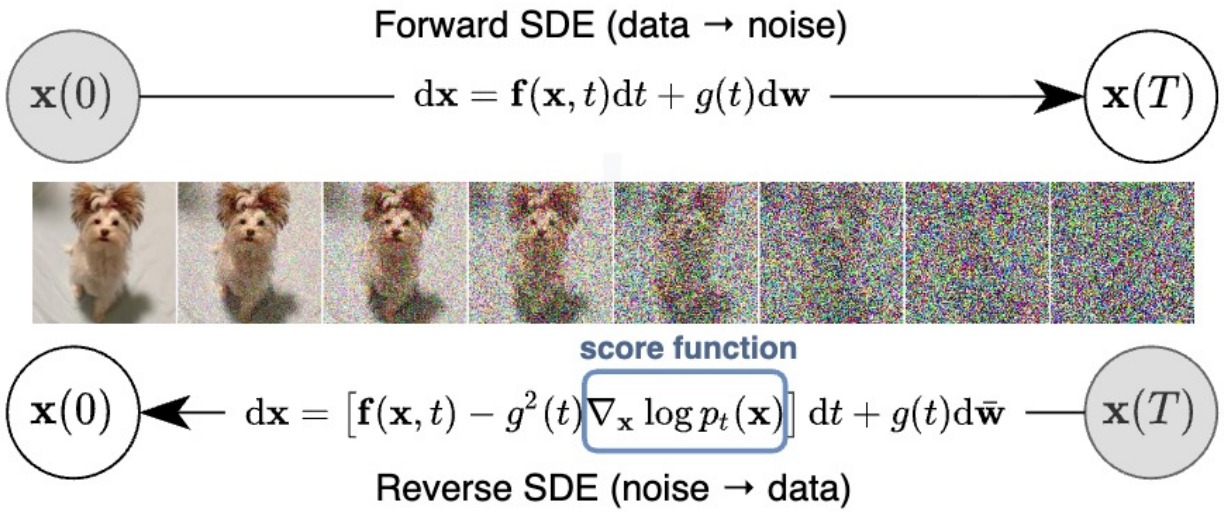
$$\mathrm{d}\mathbf{z} = \left[-\frac{1}{2}\beta(t)\mathbf{z} - \beta(t)\nabla_{\mathbf{z}}\log p_t(\mathbf{z})\right]\mathrm{d}t + \sqrt{\beta(t)}\mathrm{d}\bar{\mathbf{w}}$$

反向漂移　　得分引导　　反向注入噪声

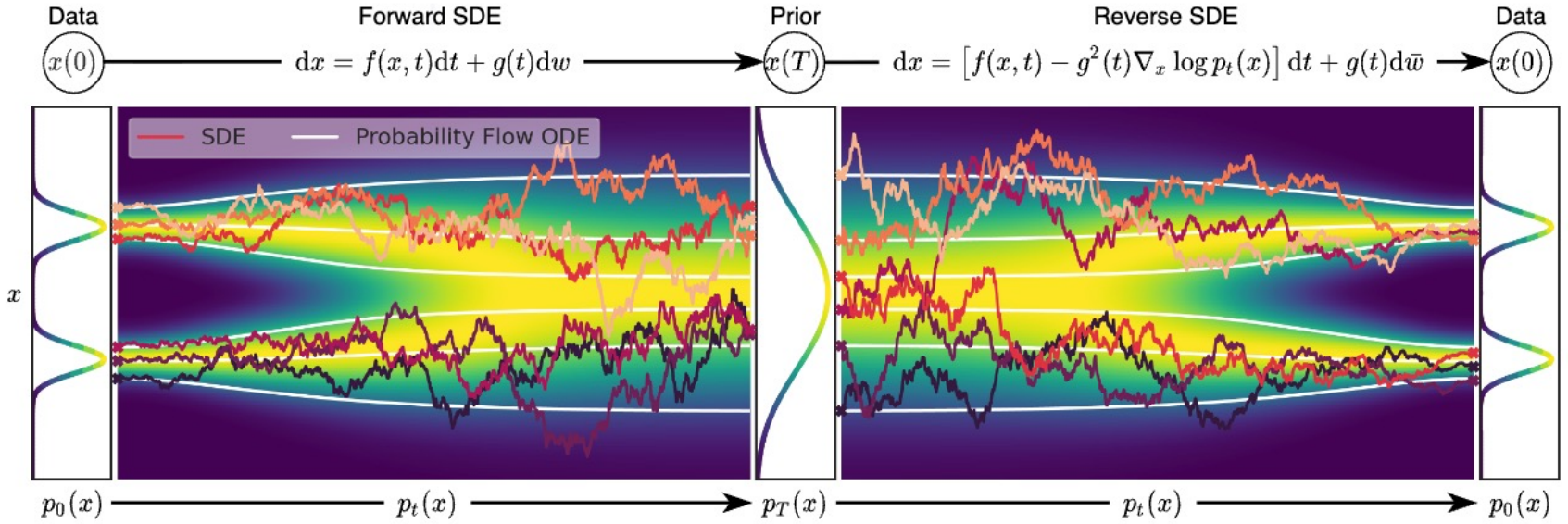
时间倒流的连续微分方程，利用连续的“得分向量场” $\nabla_z \log p_t(z)$作为引力，将粒子极其平滑地吹回真实流形。

Score-Based Generative Modeling through Stochastic Differential Equations， Yang Song et al. ICLR 2021, 杰出论文奖

随机微分方差（SDE）



https://arxiv.org/pdf/2011.13456



西南交通大学
Southwest Jiaotong University

SCAI011112 视觉计算

Thank You